The Allocation of Immigrant Talent: Macroeconomic Implications*

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Abstract

We quantify the macroeconomic costs of barriers to immigrant labor-market integration in the United States. We develop an occupational-choice model featuring natives and immigrants of multiple types who differ in productivity, preferences, and the wedges they face that distort occupational choices. Estimating the model with U.S. micro-data, we find that immigrant barriers are sizable, especially for recent, non-fluent, and undocumented immigrants. Removing all barriers raises output by 1.5%. While targeted policies yield modest gains individually, jointly reducing barriers related to legal status, language, and the integration of recent immigrants delivers gains exceeding half of those achieved by removing all barriers.

Keywords: Immigration, occupational barriers, mobility, misallocation
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1 Introduction

Immigration policy is central to economic and political debates, especially amid labor shortages and aging populations. Yet, while much attention is given to how many and whom to admit, less is devoted to how effectively immigrants integrate into labor markets and their aggregate implications. At the micro level, a large literature documents specific labor-market barriers—such as licensing, language proficiency, and discrimination—but typically studies them in isolation. At the macro level, another literature examines how immigrants affect aggregate outcomes without considering these barriers (Borjas, 1995; Llull, 2018; Monras, 2020; Albert and Monras, 2022; Monras et al., 2022). This paper bridges these two literatures by quantifying the costs of barriers to immigrant labor-market integration and the effects of policies aimed at reducing them.

Our main contribution is to combine micro-level studies on specific immigrant barriers with macro-level analysis of immigration in a unified framework that quantifies (*i*) their joint macroeconomic effects, (*ii*) how they vary across groups, and (*iii*) their implications for immigrant support policies. We develop an occupational choice model of natives and immigrants with two key features: immigrant-specific barriers that distort occupational choices and prevent immigrants from working in occupations where they are most productive, and rich heterogeneity across immigrants by legal status, time since arrival, language proficiency, and education. We apply the framework to U.S. microdata to estimate immigrant barriers across workers and occupations and to evaluate the effects of reducing them.

We highlight three key findings. First, immigrant barriers in the U.S. are sizable and vary across occupations and immigrant groups. Second, removing these barriers has large aggregate effects: lowering barriers for all immigrants to the level faced by natives raises output by 1.5 percent. Third, targeted policies addressing legal status, language proficiency, and the integration of recent immigrants generate modest gains individually. Jointly, however, these policies deliver gains exceeding half of those from removing all immigrant barriers, suggesting that immigrants often face multiple, overlapping barriers to labor-market integration.

We develop an occupational-choice model that extends the framework of Hsieh et al. (2019) by modeling immigrants as in Burstein et al. (2020), and by introducing endogenous labor supply. The model features natives and immigrants of multiple types who choose their occupation and hours worked, or to remain non-employed. Individuals of each type differ in productivity, preferences, and wedges across occupations. Each type, defined by nativity and demographics, has a common occupation-specific productivity, allowing immigrant and native productivity to differ across occupations. In addition, individuals draw idiosyncratic occupation-specific productivities from a distribution that is calibrated to capture native-immigrant differences from unobserved heterogeneity and selection. All individuals face heterogeneous preferences and occupation-specific compensation wedges modeled as taxes. Immigrants differ from natives in two ways.

First, immigrants may differ from natives in occupation-specific preferences and face additional compensation wedges, with the latter capturing labor-market barriers. Second, occupation-specific goods are produced using native and immigrant labor that are imperfect substitutes.

We derive analytical expressions to characterize the efficient allocation in our model. This benchmark corresponds to the case in which immigrant-specific wedges are eliminated and both natives and immigrants face identical distortions across occupations. In this benchmark, individuals of the same observable type should (*i*) earn the same average hourly wages in any occupation and (*ii*) have the same native-immigrant gap in average annual earnings across all occupations. These conditions provide a clear theoretical benchmark against which to assess any observed disparities in labor market outcomes between natives and immigrants across occupations.

Next, we empirically characterize these disparities by documenting key patterns of the joint distribution of employment, annual earnings, and hourly wages across individuals and occupations. Using microdata from the American Community Survey (ACS), we identify substantial differences between natives and immigrants, as well as across immigrant groups. We classify immigrants by time since arrival, English proficiency, and legal status, and further partition all individuals by education, age, and gender. We then allocate individuals between a non-market (non-employment) and 33 market occupations. Our findings reveal large deviations from the efficient benchmark because observably similar immigrants and natives exhibit large gaps in labor-market outcomes across occupations, consistent with immigrant-specific distortions.

To determine whether observed differences in immigrant labor market outcomes reflect differences in productivity or wedges, we build on insights from Hsieh et al. (2019) and develop a strategy to disentangle them. Our model links individuals' outcomes to three underlying forces: how productive they are in each occupation, how desirable different occupations are to them, and their pay once employed. This structure lets us identify productivity, immigrant-specific preferences, and immigrant compensation wedges using data on occupational allocations, earnings, and hourly wages. We begin by identifying immigrant preferences, which capture differences in the attractiveness of occupations for immigrants relative to observationally similar natives. These are recovered by comparing how immigrant earnings vary across occupations relative to natives, because conditional on wages and occupational choices, the model implies that differences in earnings are informative about utility rather than productivity or pay. Next, we identify productivity using within-immigrant variation in earnings, hourly wages, and occupational choices. More productive immigrants tend to sort into certain occupations and have higher earnings relative to their hourly wage, allowing us to infer productivity from these patterns. Finally, we back out immigrant compensation wedges by comparing hourly wages of immigrants and natives within the same occupation, after accounting for productivity and selection.

Using our identification strategy, we back out wedges from a limited set of predetermined parameters and widely available data. The model can then match the joint distribution of em-

ployment, annual earnings, and hourly wages across individuals and occupations. The estimated wedges and native-immigrant productivity differences are sizable and vary across immigrant types and occupations. For instance, recent, low-English, and undocumented immigrants tend to be more productive than natives in manual occupations, yet also face the largest wedges there.

A potential concern is that the estimated wedges capture persistent unobserved productivity differences rather than labor-market barriers. We address this concern through validation exercises to examine whether the wedges vary in ways consistent with barriers rather than persistent unobserved productivity differences. First, we document that the wedges decline monotonically with time since arrival, consistent with barriers that are important at entry but weaken with labor-market integration, rather than productivity differences. Second, we validate the wedges against external occupation-level measures. Their relationship with licensing requirements is strongest for recent and undocumented immigrants, consistent with credentialing barriers that are especially important upon entry and for workers without status, and that weaken with labor-market integration over time. Their relationship with occupational language intensity is strongly positive for non-fluent immigrants but weak for fluent immigrants, consistent with language-intensive occupations imposing larger barriers on the group for whom English is most limiting. If the wedges mainly reflected persistent unobserved productivity differences, there would be little reason to expect them to decline with time since arrival, for their relationship with licensing requirements to be strongest for recent and undocumented immigrants, or for them to be primarily associated with language intensity among non-fluent immigrants.

After validating the wedges, we analyze the macroeconomic and welfare effects of reducing them. First, we set all immigrant wedges equal to those of natives. Next, we study three targeted counterfactuals tied to policy-relevant margins: legal status, language proficiency, and the labor-market integration of recent arrivals. These margins correspond to policies such as access to legal work authorization, language training, credential recognition, and job-search assistance. Finally, we examine the three counterfactuals jointly to assess the gains from addressing them together.

The benchmark policy of removing all immigrant wedges yields a 1.5 percent increase in output. This should be viewed as an upper bound, as it is hard to conceive of policies that remove *all* immigrant barriers. Individually, the targeted policies yield more modest gains. The legal-status and language counterfactuals generate gains largely in routine manual jobs, reallocate workers from non-employment into manual jobs, and reallocate some incumbent workers into cognitive jobs. These counterfactuals disproportionately affect vulnerable groups, such as less-educated, undocumented, and immigrants with limited English proficiency. By contrast, targeting recent immigrants generates gains primarily in non-routine cognitive jobs and reallocates more educated workers. Critically, while individual policies yield modest gains, the joint policy generates gains exceeding half those of the upper-bound benchmark. These larger gains arise because barriers are correlated, so targeting them individually leaves other barriers in place. For example,

undocumented immigrants are also more likely to face difficulties in language proficiency and credential acquisition. In practice, this finding supports a more coordinated approach to immigrant integration. These patterns also carry over to the welfare implications of immigrant barriers. Welfare gains are largest for groups exposed to multiple barriers, and the joint counterfactual delivers the largest welfare gains for most immigrant subgroups.

Related literature. Our paper contributes to a growing literature that studies the aggregate effects of immigration using structural frameworks (Llull 2018; Burstein et al. 2020; Monras 2020; Albert 2021; Albert et al. 2021; Hanson and Liu 2021; Piyapromdee 2021; Albert and Monras 2022; Mandelman and Zlate 2022; Monras et al., 2022). These papers develop models disciplined by microdata to analyze the impact of immigration on wages, migration, inequality, polarization, output, and welfare. We contribute to this literature by documenting occupational differences in labor-market outcomes between natives and immigrants, and by using these patterns to estimate immigrant barriers and evaluate their macroeconomic and policy implications.

Another literature studies immigrant outcomes in the context of cross-country differences in human capital and productivity (Hendricks 2002; Schoellman 2012; Schoellman 2016; Hendricks and Schoellman 2018; Lagakos et al. 2018a; Martellini et al. 2023). Our findings speak to this literature by showing that immigrant barriers can reduce their productivity. Although we focus on the U.S., this mechanism applies to other countries with growing immigrant labor forces.

A separate literature documents that immigrants have a disadvantage in labor markets due to occupational regulations and licensing (Peterson et al. 2014), lower bargaining power against employers (Moreno-Galbis and Tritah 2016), discrimination (Oreopoulos 2011), and occupational and skill downgrading (Eckstein and Weiss 2004; Dustmann et al. 2013; Lessem and Sanders 2020). These barriers lead to immigrants' poorer labor market outcomes (Abramitzky and Boustan 2017; Arellano-Bover and San 2020; Dostie et al. 2020). Generally, most work in this literature explores one aspect of labor market barriers. Relative to the existing literature, our paper recovers the joint macroeconomic effect of immigrant barriers.

Finally, our paper also contributes to a literature on the macroeconomic effects of the misallocation of factor inputs across production units, sectors, and occupations (Restuccia and Rogerson 2008; Hsieh and Klenow 2009; Buera et al. 2011; Bartelsman et al. 2013; Hopenhayn 2014; Bento and Restuccia 2017; Gopinath et al. 2017; Hsieh et al. 2019). Relative to this body of work, we focus on the misallocation of immigrants, which represents a substantial share of employment in the United States. We show that immigrants face substantial wedges that distort their labor supply decisions, with significant implications for aggregate outcomes.

The paper proceeds as follows. Section 2 presents the model. Section 3 details the data, estimation strategy, and identification. Section 4 reports estimation results and our validation exercises. Section 5 shows the aggregate and heterogeneous effects of reducing immigrant wedges. Section 6 examines robustness under alternative model specifications. Section 7 concludes.

2 Model

In this section, we develop a static occupational choice model à la Roy (1951) featuring natives and immigrants of multiple types. This framework extends the model in Hsieh et al. (2019) by incorporating immigrants as in Burstein et al. (2020), and by endogenizing labor supply. Relative to the former, our model lacks dynamics; relative to the latter, it distinguishes immigrants by language ability, legal status, time since arrival, and demographics, but not by region.

We consider an economy populated by a continuum of individuals and a discrete number of occupations. Individuals choose their occupation and hours worked, and production in each occupation is carried out by a representative firm that hires labor. A representative final-good producer aggregates the production from each occupation into a final good. We first describe the frictionless environment, which provides a benchmark in which allocations between natives and immigrants can differ only because of differences in productivity and preferences across occupations, and then extend it to allow for immigrant barriers.

2.1 Individuals

Demographics. We consider a static model in which individuals live for one period. They are partitioned into types $i = 1, ..., I$ based on their immigration status (e.g., natives and various types of immigrants based on time since immigration, English fluency, and legal status). We let $i = 1$ denote natives and $i = 2, ..., I$ denote the set of immigrant types. Individuals of every given type i are further partitioned into subtypes $g = 1, ..., G$ based on observables such as age, gender, and education. We denote the mass of individuals of type i and subtype g by N_{ig}; the total mass of individuals in the economy is N, $\sum_{i=1}^{I} \sum_{g=1}^{G} N_{ig} = N$.

Preferences. Individuals of type i and subtype g supply ℓ units of labor in occupation $j = 0, \ldots, J$ and consume c units of the final good. Their preferences over consumption and labor supply are represented by the following utility function: $u_{ig}^{j}(c, \ell) = (1 + \gamma_{ig}^{j})\nu_{g}^{j} c - \frac{\ell^{1+\frac{1}{\xi}}}{1+\frac{1}{\xi}}$, where ξ is the Frisch elasticity, ν_{g}^{j} is a preference shifter common to all individuals of subtype g in occupation j, and γ_{ig}^{j} is an immigrant-specific preference shifter common to all individuals of type i and subtype g in occupation j. Thus, $\gamma_{1g}^{j} = 0$ for all g, j, as $i = 1$ denotes natives.

Individual productivity across occupations. The supply of labor by individuals is not equally productive in all occupations. An individual of type i and subtype g who chooses to supply ℓ units of labor to work in occupation j supplies $z_{ig}^{j} \varepsilon_{j} \ell$ effective units of labor, where z_{ig}^{j} is a productivity component common across all individuals of type i and subtype g that work in occupation j, while ε_{j} is an idiosyncratic occupation-specific productivity draw.

Each individual of type i and subtype g is characterized by a vector of idiosyncratic productivities $(\varepsilon_0, ..., \varepsilon_J)$ for each of the occupations. These productivities are distributed Fréchet with type-specific shape parameter η_i and *i.i.d.* across individuals and occupations, as in Mc-

Fadden (1972), Eaton and Kortum (2002), and Hsieh et al. (2019). The joint CDF is thus given by $F(\varepsilon_0, ..., \varepsilon_j) = \exp\left(-\sum_{j=0}^{J} \varepsilon_j^{-\eta_i}\right)$. We model η_i as type-specific to capture other sources of productivity differences between natives and immigrants that are not modeled explicitly. These sources could include selection (Hendricks and Schoellman, 2018) or unobserved heterogeneity (e.g., due to origin-country differences in education quality Martellini et al., 2023 or in life-cycle human capital accumulation Lagakos et al., 2018b) across types.

Labor income and occupational choice. Individuals of type i and subtype g who work in occupation j are paid a wage w_{ig}^j per effective unit of labor. An individual with a vector of idiosyncratic productivities $(\varepsilon_0, ..., \varepsilon_J)$ chooses occupation j^* and labor supply ℓ^* that solve:

$$\max_{j \in \{0,...,J\}, \ell} (1 + \gamma_{ig}^j) \nu_g^j c - \frac{\ell^{1+\frac{1}{\xi}}}{1+\frac{1}{\xi}} \quad s.t. \quad pc = w_{ig}^j \ell z_{ig}^j \varepsilon_j,$$

where p denotes the price of the final good.

2.2 Occupations

Production in each occupation $j = 0, ..., J$ is carried out by an occupation-specific representative firm. Occupation $j = 0$ is the non-market occupation (i.e., work at home as in Hsieh et al. 2019), while the rest, $j = 1, ..., J$, are market occupations.

We model the difference between market and non-market occupations by assuming that they differ in their production technologies. Production in market occupations is carried out through a nested constant elasticity of substitution (CES) technology that aggregates different types of labor to produce an occupation-specific good. In contrast, production in the non-market occupation is carried out through a linear technology, capturing the idea that this occupation encompasses home production activities that could be done independently by each individual.

2.2.1 Market occupations

Following Burstein et al. (2020), production in each market occupation is described by a nested CES technology. The outer nest combines native and immigrant labor bundles, while the inner nests aggregate labor across worker types and demographic subtypes within each of these two bundles. We describe the two nests in turn.

Outer nest: Aggregation between natives and immigrants. The production technology for the outer nest aggregates labor bundles between natives and immigrants with a CES technology with elasticity σ_j: $y_j = A_j[n_{\text{nat}}^{j\ \frac{\sigma_j-1}{\sigma_j}} + n_{\text{imm}}^{j\ \frac{\sigma_j-1}{\sigma_j}}]^{\frac{\sigma_j}{\sigma_j-1}}$, where y_j is the output produced in occupation j, n_k^j is the labor bundle of group k in occupation j, and A_j is occupation-specific productivity. We index natives and immigrants with subscripts $k = \text{nat}$ and $k = \text{imm}$.[1]

[1]We also study two alternative production technologies. In the first, the outer nest aggregates natives and immigrants across different education levels. Specifically, the outer nest aggregates labor bundles across natives with a college degree, natives without a college degree, immigrants with a college degree, and immigrants without

The problem of the representative producer in market occupation $j = 1, ..., J$ involves maximizing profits by choosing the amount of labor bundles of each group to hire, taking as given the price of the good sold and the wage rate of each labor bundle. This problem is given by:

$$\max_{y_j, n^j_{\text{nat}}, n^j_{\text{imm}}} p_j y_j - w^j_{\text{nat}} n^j_{\text{nat}} - w^j_{\text{imm}} n^j_{\text{imm}} \quad s.t. \quad y_j = A_j \left[{n^j_{\text{nat}}}^{\frac{\sigma_j - 1}{\sigma_j}} + {n^j_{\text{imm}}}^{\frac{\sigma_j - 1}{\sigma_j}} \right]^{\frac{\sigma_j}{\sigma_j - 1}},$$

where p_j denotes the price of the good produced by occupation j, and w^j_k denotes the cost of labor bundle k hired by occupation j.

Inner nest: Aggregation within natives and immigrants. The production technology for the inner nest produces labor bundles for group $k \in \{\text{nat,imm}\}$ by aggregating workers of all types $i \in \mathcal{I}_k$ and all subtypes g with a CES technology with elasticity $\widetilde{\sigma}_j$ for each $k \in \{\text{nat, imm}\}$: $n^j_k = [\sum_{i \in \mathcal{I}_k} \sum_{g=1}^{G} {n^j_{ig}}^{\frac{\widetilde{\sigma}_j - 1}{\widetilde{\sigma}_j}}]^{\frac{\widetilde{\sigma}_j}{\widetilde{\sigma}_j - 1}}$, where $\mathcal{I}_{\text{nat}} = \{1\}$, $\mathcal{I}_{\text{imm}} = \{2, ..., I\}$, and n^j_{ig} denotes the effective units of labor hired from individuals of pair (i, g) in occupation j.

The problem of the representative producer of labor bundles of group $k \in \{\text{nat, imm}\}$ in market occupation $j = 1, ..., J$ consists of maximizing profits by choosing the total effective units of labor of each type and subtype to hire, taking as given the price of the labor bundle and wage rates in occupation j. This problem is given by:

$$\max_{n^j_k, \{n^j_{ig}\}_{i \in \mathcal{I}_k, g}} w^j_k n^j_k - \sum_{i \in \mathcal{I}_k} \sum_{g=1}^{G} w^j_{ig} n^j_{ig} \quad s.t. \quad n^j_k = \left[\sum_{i \in \mathcal{I}_k} \sum_{g=1}^{G} {n^j_{ig}}^{\frac{\widetilde{\sigma}_j - 1}{\widetilde{\sigma}_j}} \right]^{\frac{\widetilde{\sigma}_j}{\widetilde{\sigma}_j - 1}},$$

where w^j_{ig} is the wage rate per effective unit of labor for pair (i, g) in occupation j.

2.2.2 Non-market occupation

Production in the non-market occupation $j = 0$ is carried out by a representative firm using labor of any type and subtype. The production technology is linear in the total effective units of labor with occupation-specific productivity A_0. The problem of this firm consists of maximizing profits by choosing the total effective units of labor hired n^0 given the price of the good sold p_0 as well as the occupation-specific wage rate w^0. This problem is given by:

$$\max_{y_0, n^0} p_0 y_0 - w^0 n^0 \quad s.t. \quad y_0 = A_0 n^0.$$

2.3 Final good producer

The final good is produced by a representative firm that aggregates the goods produced across all occupations by operating a CES technology with elasticity σ.

The problem of the final-good producer consists of maximizing profits by choosing the amount of goods to purchase from each of the occupations y_j, taking as given the price of the final good

a college degree. In the second, the outer nest aggregates labor bundles across natives and each different type of immigrant. In Section 6, we discuss the implications of these alternative specifications.

p as well as prices of occupation-specific goods p_j. This problem is given by:

$$\max_{y,\{y_j\}_{j=0}^J} py - \sum_{j=0}^{J} p_j y_j \quad s.t. \quad y = \left[\sum_{j=0}^{J} y_j^{\frac{\sigma-1}{\sigma}}\right]^{\frac{\sigma}{\sigma-1}}.$$

2.4 Equilibrium

The equilibrium consists of prices and allocations such that (i) agents solve their problem taking prices as given; (ii) labor markets for each (type, subtype) pair in each occupation clear; and (iii) the final good market clears. We provide a formal definition in Appendix A.

2.5 Immigrant barriers

The environment described above allows natives and immigrants to differ across occupations in their productivity and preferences. Yet, in an occupation, the compensation that individuals receive may also reflect barriers or distortions unrelated to their productivity and preferences. We now extend the environment to capture such differences, which we model as compensation wedges. These wedges have two components: one that is common to all individuals in a given occupation, and one that is specific to immigrants and varies across immigrant types.

Compensation wedges. We model compensation wedges as proportional taxes (or subsidies) on labor income. All individuals of subtype g that work in occupation j are subject to a common compensation wedge τ_g^j. Immigrants of type $i = 2, ..., I$ are additionally subject to "immigrant compensation wedges" κ_{ig}^j. Thus, $\kappa_{1g}^j = 0 \ \forall g, j$ since $i = 1$ denotes natives. We assume that all revenue collected from these wedges is rebated to all individuals as a proportional subsidy s.

The immigrant compensation wedge κ_{ig}^j is the central object of our analysis. A positive κ_{ig}^j implies that immigrants of type i and subtype g receive lower compensation per effective unit of labor than natives of the same subtype g in occupation j, after accounting for differences in productivity and preferences. We interpret κ_{ig}^j as a reduced-form wedge that captures barriers immigrants face within an occupation. These barriers may operate on either side of the labor market: through constraints that immigrants themselves face or through employers' reluctance to hire or promote them as readily as comparable natives. Such barriers are distinct from policy levers that govern who is admitted into the country at the border and in what numbers: rather than shaping the size or composition of immigrant inflows, κ_{ig}^j captures how effectively immigrants who are already in the country can deploy their labor within the host-country market. The common wedge τ_g^j, by contrast, applies to natives and immigrants alike and absorbs occupation-level distortions that are not specific to immigrant status.

Because κ_{ig}^j varies across immigrant types, it can be mapped to specific barriers. We focus on three margins along which policy can plausibly act; in each, the barrier appears as a larger wedge for the more disadvantaged group, and the policy we consider in Section 5 brings that

group's wedge down toward that of its better-positioned counterpart. The first is legal status: undocumented immigrants are barred from formal positions and have little recourse against under-compensation, so they tend to earn less than comparable documented immigrants. The corresponding policy is an amnesty that regularizes their status. The second is English fluency: immigrants with limited host-country language proficiency are less able to enter and be fully compensated in communication-intensive roles relative to otherwise similar fluent immigrants. Here, policy can act through language training and the building of workplace-relevant communication skills. The third is time since arrival: recent arrivals have yet to accumulate the host-country knowledge and professional networks that open access to better-paying jobs, so their barriers are largest on arrival and recede with tenure. While our model is static, this process is consistent with dynamic models in which immigrants accumulate human capital after arrival (e.g., Adda et al., 2022). Relevant policies aim to accelerate this integration, such as job-search and placement assistance, mentoring and professional-networking programs, and orientation programs that connect recent arrivals to host-country employers. Finally, as a benchmark, we also consider removing immigrant compensation wedges entirely, which bounds the potential gains.

Occupational choice. Under the extended environment, an individual solves:

$$\max_{j\in\{0,...,J\},\ell} (1+\gamma_{ig}^{j})\nu_{g}^{j}c - \frac{\ell^{1+\frac{1}{\xi}}}{1+\frac{1}{\xi}} \quad s.t. \quad pc = (1-\tau_{g}^{j}-\kappa_{ig}^{j})w_{ig}^{j}\ell z_{ig}^{j}\varepsilon_{j} \times (1+s), \tag{1}$$

where the right-hand side of the budget constraint is individual labor income net of compensation wedges τ_g^j and κ_{ig}^j, along with reimbursement s.

Equilibrium. The equilibrium definition extends the one above by adding the condition that the aggregate revenue collected through compensation wedges equals aggregate reimbursements.

3 Estimation

3.1 Data

We estimate the model using U.S. data from the American Community Survey (ACS) between 2010 and 2019.[2] We restrict our sample to non-business owners between the ages of 25 and 54. This sample restriction allows us to focus on working-age individuals who have finished schooling but are prior to retirement. We also drop individuals on active military duty. Appendix B.1 provides details about the data, construction of variables, and measurement.

Individual types. We partition individuals in the data into the I types outlined in the model, which we index by $i = 1, ..., I$. We let $i = 1$ denote the set of natives and let $i = 2, ..., I$ denote the partition of immigrants based on time since immigration (tenure), English fluency, and legal

[2]We pool all ten years together to increase the sample size and treat them as one cross section.

status. We define immigrants as the set of foreign-born individuals.[3]

We partition immigrants by years since arrival in the U.S. into three groups: recent immigrants (0–4 years), intermediate immigrants (5–10 years), and established immigrants (>10 years).[4] We classify immigrants' English proficiency based on respondents' self-reported assessment in the ACS. In particular, we distinguish between two groups: fluent immigrants, who speak only English or speak very well, non-fluent immigrants, who do not speak English very well. Finally, we also partition immigrants by legal status into documented and undocumented immigrants. Details on how we identify undocumented immigrants are discussed below.

Thus, we consider an economy with 13 individual types ($I = 13$): one type for natives and 12 types for immigrants, namely 3 (time since arrival) $\times$ 2 (English fluency) $\times$ 2 (legal status).

Individual subtypes. We further partition each type $i = 1, \ldots, I$ into G subtypes, indexed by $g = 1, \ldots, G$, based on gender, education, and age. We distinguish two gender groups (male and female), four education groups (less than high school, high school, some college, and college or more), and three age groups (25–34, 35–44, and 45–54). This yields 24 subtypes ($G = 24$) per type, implying a total of 312 worker (type, subtype) pairs in the data.

Identifying undocumented immigrants. Because the ACS does not directly report immigrants' legal status, we closely follow the imputation procedure in Borjas (2017) to categorize immigrants as either documented or undocumented. Specifically, we classify an immigrant as documented if *any* of the following conditions holds: the person arrived in the U.S. before 1980; is a U.S. citizen; receives Social Security, Supplemental Security Income (SSI), Medicaid, Medicare, or military insurance; is a veteran or is currently serving in the armed forces; works in the public sector; was born in Cuba; is employed in a licensed occupation; or has a spouse who is a legal immigrant or citizen. Relative to Borjas (2017), the only criterion in his classification that we are unable to reproduce in the ACS is residence in public housing or receipt of rental subsidies. Thus, our implementation slightly differs from Borjas's because his procedure is applied to CPS ASEC data with those additional information, whereas ours is applied to the ACS.

Table 1 reports the demographic composition of documented and undocumented immigrants. Undocumented immigrants are more likely than their documented counterparts to be male, less educated, younger, recent arrivals, and less fluent in English. Across broad occupation groups, they are more likely to be non-employed (defined below) or to work in manual occupations.

Overall, we estimate that 36 percent of immigrants are undocumented. While this estimate is somewhat high, it is still broadly comparable to existing estimates. For example, Passel and Krogstad (2025) estimate that unauthorized immigrants accounted for 27 percent of the U.S.

[3]Specifically, the group of immigrants includes naturalized citizens and non-citizens. However, we classify natives' foreign-born children as natives.

[4]The five-year cutoff is motivated in part by U.S. immigration institutions: five years of continuous residence as a lawful permanent resident is the standard residence requirement for naturalization.

Table 1: Demographic composition of documented and undocumented immigrants

Group	Documented	Undocumented
Population share among immigrants	0.64	0.36
Male	0.46	0.54
Female	0.54	0.46
No High School	0.20	0.39
High School	0.22	0.23
Some college	0.23	0.13
College	0.35	0.26
Age 25–34	0.27	0.41
Age 35–44	0.37	0.37
Age 45–54	0.37	0.22
Recent (0–4 years)	0.08	0.21
Intermediate (5–10 years)	0.14	0.24
Established (10+ years)	0.78	0.55
Fluent English	0.59	0.35
Non-fluent English	0.41	0.65
Non-routine cognitive	0.28	0.16
Non-routine manual	0.14	0.19
Routine cognitive	0.13	0.08
Routine manual	0.17	0.27
Non-market	0.27	0.30

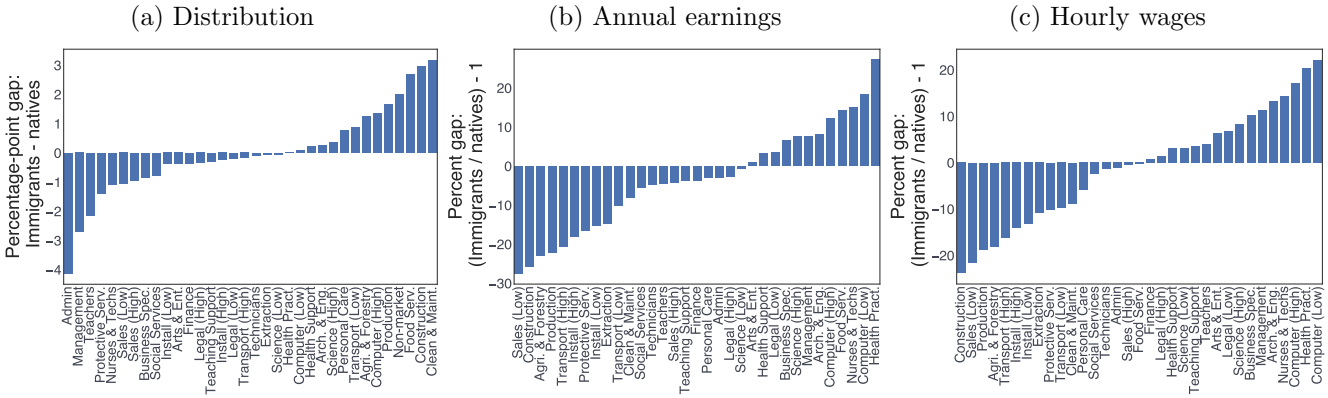
Notes: This table reports the demographic composition of documented and undocumented immigrants in the ACS. The first row reports each group's population share among all immigrants. All remaining rows report within-group shares, so that the entries within each demographic block sum to one for documented immigrants and for undocumented immigrants separately, up to rounding. The blocks correspond to gender, education, age, time since arrival (tenure), English proficiency, and broad occupation group. The classification of immigrants into documented and undocumented status follows the imputation procedure in Borjas (2017). Occupation groups are defined as non-routine cognitive, non-routine manual, routine cognitive, routine manual, and non-market (see Table A1).

foreign-born population in 2023. While we try to adhere as closely as possible to Borjas (2017), Section 6 considers a less restrictive version of the classification rule for assigning immigrants to documented status. Specifically, we expand the documented category to include immigrants who receive any welfare benefits or are current graduate students. Under this alternative classification, the fraction of undocumented immigrants falls to 28.7 percent of immigrants, even closer to available external estimates. Section 6 shows that our main results remain similar in this case.

Market vs. non-market occupations. We allocate individuals between non-market and market occupations. We classify an individual as being in the non-market occupation if she is not employed or employed but usually works less than 10 hours per week. An employed individual who usually works at least 10 hours is assigned to one of the market occupations.

Market occupations. Our grouping of market occupations begins from the two-digit 2010 Standard Occupational Classification (SOC) system, as recorded in the ACS. We then refine these categories to capture meaningful heterogeneity within broad SOC groups, arriving at 33 market occupations ($J = 33$), listed in Table A1; Appendix B.1 provides details. We split some

Figure 1: Immigration distribution, earnings, and wages across occupations relative to natives



Notes: This figure plots the distribution, annual earnings, and hourly wages of immigrants across occupations relative to those of natives using data from the 2010-2019 ACS. We calculate the fraction of immigrants (natives) in each occupation among all immigrants (natives). Panel (a) shows the percentage-point gap (calculated as immigrants − natives) between fractions of immigrants and natives. Panels (b) and (c) show the percent gap (calculated as immigrants/natives − 1) between their annual earnings and hourly wages.

occupations into finer groups when there is meaningful heterogeneity in tasks, skill requirements, managerial and supervisory responsibilities, or institutional barriers within the underlying SOC category. In particular, we define "high" and "low" subgroups where appropriate. For example, we split computer-related occupations into a higher-skill group of computer scientists and programmers and a lower-skill group of other analysts and administrators. Likewise, we split legal occupations into lawyers and judges versus legal clerks and assistants.

Annual earnings and hourly wages. We measure the annual earnings as total annual labor income (in 2019 dollars). We also measure hourly wages of individuals as the ratio of annual earnings to total annual hours worked. For each set of individuals of type i and subtype g in market occupation j, we compute the group's average annual earnings and average hourly wages as a geometric average among employed individuals with non-missing labor earnings information.

Summary statistics. As seen in Figure 1, immigrants are unevenly distributed across occupations, with overrepresentation in several manual and service occupations and underrepresentation in many managerial, administrative, and teaching-related occupations. The share of non-employed individuals (i.e., those in the non-market occupation) is 2 percentage points higher for immigrants than for natives. The earnings and wage gaps also vary substantially across occupations but are highly correlated with each other in that occupations with the largest wage gaps tend to also have the largest earnings gaps. This indicates that the earnings gap is primarily due to differences in hourly wages rather than differences in hours.

Table 2 shows that recent immigrants are more likely to be non-employed, but the gap relative to natives narrows over time and disappears for established immigrants. Non-fluent and undocumented immigrants are also more likely to be non-employed. Fluent immigrants tend to work in non-routine cognitive occupations, whereas non-fluent immigrants sort into manual ones. Undocumented immigrants are also much more likely to work in manual jobs than documented

Table 2: Empirical moments

	Distribution							
Occupation type	N	I_{0-4}	I_{5-10}	I_{10+}	$I_{\text{Low Eng}}$	$I_{\text{High Eng}}$	I_{Undoc}	I_{Doc}
Non-routine cognitive	0.31	0.25	0.23	0.24	0.09	0.38	0.16	0.28
Non-routine manual	0.11	0.13	0.18	0.16	0.21	0.11	0.19	0.14
Routine cognitive	0.18	0.08	0.10	0.12	0.08	0.14	0.08	0.13
Routine manual	0.15	0.15	0.20	0.22	0.29	0.13	0.27	0.17
Non-market	0.26	0.39	0.29	0.26	0.33	0.23	0.30	0.27
	Annual earnings							
Occupation type	N	I_{0-4}	I_{5-10}	I_{10+}	$I_{\text{Low Eng}}$	$I_{\text{High Eng}}$	I_{Undoc}	I_{Doc}
Non-routine cognitive	1.79	1.69	1.94	2.14	1.52	2.17	2.11	2.03
Non-routine manual	0.76	0.46	0.55	0.66	0.55	0.73	0.55	0.66
Routine cognitive	1.04	0.71	0.80	0.98	0.73	1.04	0.82	0.97
Routine manual	1.08	0.61	0.72	0.89	0.75	1.02	0.73	0.93
	Hourly wages							
Occupation type	N	I_{0-4}	I_{5-10}	I_{10+}	$I_{\text{Low Eng}}$	$I_{\text{High Eng}}$	I_{Undoc}	I_{Doc}
Non-routine cognitive	1.85	1.94	2.08	2.23	1.70	2.27	2.24	2.14
Non-routine manual	0.92	0.68	0.71	0.81	0.71	0.90	0.69	0.84
Routine cognitive	1.18	0.96	0.97	1.12	0.89	1.20	0.97	1.13
Routine manual	1.17	0.79	0.82	0.98	0.86	1.11	0.83	1.03

Notes: This table presents the distribution of individuals across market and non-market occupations and their associated annual earnings and hourly wages using data from the 2010-2019 ACS. We first calculate the outcomes for each individual (type, subtype) pair in each occupation. For expositional purposes, we report the average moments for natives and immigrant types across four broad occupation categories, where we assign 33 market occupations into categories based on their skill and task-intensity: non-routine cognitive, non-routine manual, routine cognitive, and routine manual. The distribution of individuals across occupations is conditional on each worker type. Annual earnings and hourly wages are expressed relative to respective values for native males of ages 25 to 34 without high school degree and employed in management occupations. N denotes natives, I_{0-4} denotes recent immigrants (≤ 4 years), I_{5-10} denotes intermediate immigrants (5–10 years), I_{10+} denotes established immigrants (> 10 years), $I_{\text{Low Eng}}$ denotes low English proficiency immigrants, $I_{\text{High Eng}}$ denotes high English proficiency immigrants, I_{Undoc} denotes undocumented immigrants, and I_{Doc} denotes documented immigrants.

immigrants. Regarding earnings, recent, non-fluent, and undocumented immigrants tend to earn less relative to natives. However, these gaps are much smaller in non-routine cognitive occupations. Established, fluent, and documented immigrants earn amounts that are closer to those of natives or, in the case of non-routine cognitive occupations, even higher.[5]

3.2 Estimation approach

We now present our approach to estimating model parameters. The parameter space is partitioned into two groups. The first is predetermined and set externally. The second is estimated to match features of the data. Table 3 summarizes our estimation approach, listing the predetermined and estimated parameters and the moments used to pin down the latter.

The set of predetermined parameters consists of ξ, $\{\eta_i\}_{i=1}^{I}$, σ, $\{\sigma_j\}_{j=1}^{J}$, and $\{\widetilde{\sigma}_j\}_{j=1}^{J}$. We set the Frisch elasticity $\xi = 0.75$. As discussed in Section 2, the shape parameter η_i of the Frechet distribution of idiosyncratic productivities may vary across types to capture potential

[5]Table A2 in Appendix B.2 also reports the distribution, annual earnings, and hourly wages of immigrants across occupations when they are grouped by demographic characteristics such as age, education, and gender.

Table 3: Estimation approach: Parameters and targets

Predetermined parameters

Parameter	Value	Description
ξ	0.75	Frisch elasticity
$\{\eta_i\}_{i=1}^{I}$	4	Frechet shape
σ	20	Elasticity across sectoral goods
$\{\sigma_j\}_{j=1}^{J}$	20	Elasticity across worker bundles between natives and immigrants
$\{\widetilde{\sigma}_j\}_{j=1}^{J}$	40	Elasticity across worker bundles between individual types and subtypes

Estimated parameters

Parameter	# of parameters	Description	Normalization
$\{z_{ig}^j\}$	10,607	Individual productivity	$z_{bm}^0 = 1$
$\{\tau_g^j\}$	759	Compensation wedges	$\tau_m^j = 0\ \forall j$, $\tau_g^0 = 0\ \forall g$
$\{\kappa_{ig}^j\}$	9,504	Immigrant compensation wedges	$\kappa_{1g}^j = 0\ \forall g, j$, $\kappa_{ig}^0 = 0\ \forall i, g$
$\{\nu_g^j\}$	792	Preferences	$\nu_g^0 = 1\ \forall g$
$\{\gamma_{ig}^j\}$	9,504	Immigrant labor supply preferences	$\gamma_{1g}^j = 0\ \forall g, j$, $\gamma_{ig}^0 = 0\ \forall i, g$
$\{N_{ig}\}$	311	Mass of individuals	$\sum_{i,g} N_{ig} = 1$
$\{A_j\}$	33	Occupation productivity	$A_1 = 1$
Total	31,510		

Target moments

Moment	# of moments
Share of individuals (i, g) that work in occupation $j\ \forall i, g, j$	10,607
Avg. annual earnings of (i, g) in j relative to (b, m) in occupation 1 $\forall i, g, j$	10,607
Avg. hourly wage of (i, g) in j relative to (b, m) in occupation 1 $\forall i, g, j$	10,296
Total	35,110

Notes: Individuals of type b and subtype m are defined as the base group relative to which various parameters are normalized.

productivity differences across natives and immigrants due to unobserved heterogeneity and immigrant selection. In our baseline calibration, we assume that the idiosyncratic productivity of natives and immigrants is drawn from a common Frechet distribution with shape η—we set this value to 4, as in Hsieh et al. (2019). Importantly, we argue that this assumption is the empirically relevant case given the evidence in Martellini et al. (2023), which we discuss using Figure A1 in Appendix C.2. We nevertheless recompute our main results in Section 6 under the alternative assumption that immigrants and natives differ in their underlying productivities.

We set $\sigma_j = \sigma \quad \forall j = 1, \ldots, J$ to simplify the estimation, as this restriction allows us to analytically back out the model's parameters from the target moments. In the baseline, we set the elasticity of substitution between natives and immigrants in the outer nest to 20, following Ottaviano and Peri (2012).[6] For the inner nest, we approximate perfect substitution within natives and immigrants by setting $\widetilde{\sigma}_j = 40\ \ \forall j$, as in Burstein et al. (2020), who assume within-occupation perfect substitution (across education groups) separately for natives and immigrants.[7]

[6] Ottaviano and Peri (2012) estimate the native-immigrant elasticity of substitution within education-experience cells from a nested-CES labor-demand framework. Their estimate of 20 is obtained under the restriction that this elasticity is common across education groups, which is the closest counterpart to our baseline specification. Although their model differs from ours in other dimensions, the parameter is derived from a production structure that disciplines the same economic object of native-immigrant labor substitutability.

[7] Appendix E presents our main results under an even higher value of $\widetilde{\sigma}_j$ to approximate perfect substitution.

We recognize that estimates of substitutability vary across studies and specifications. Section 6 considers several alternatives: lower substitutability between natives and immigrants; a richer outer-nest structure in which natives and all 12 immigrant types are aggregated as separate labor bundles; an education-based nesting structure with college and non-college native and immigrant labor bundles; an occupation-varying outer-nest elasticity with lower substitutability in manual jobs; perfect substitution in the outer nest; and a higher value for the inner-nest elasticity. Taken together, these exercises show that the effects of the policy counterfactuals are qualitatively robust to a wide range of assumptions on substitutability. Quantitatively, lower substitutability in the outer nest naturally leads to larger gains from reducing immigrant wedges.

Our first step in pinning down the estimated parameters is to impose a set of normalizations and identifying assumptions. We define a base individual (type, subtype) pair, indexed by $b \in \{1, \ldots, I\}$ and $m \in \{1, \ldots, G\}$. Our first normalization is setting $z^0_{bm} = 1$. This implies that the productivity of all other individuals is expressed relative to the productivity of the base type-subtype pair (b, m) in the non-market occupation. Second, we assume all types and subtypes face no compensation wedges in the non-market occupation: $\tau^0_g = 0$ and $\kappa^0_{ig} = 0$ for all i, g. We also assume that natives belonging to the base type and subtype (b, m) face no compensation wedges in any market occupation: $\tau^j_m = 0$ for all j. Third, we normalize the common preference shifter for the non-market occupation such that $\nu^0_g = 1$ for all g. Fourth, we set the immigrant-specific preference shifter in the non-market occupation to zero: $\gamma^0_{ig} = 0$ for all i, g. Fifth, we normalize the total mass N of all individuals to be 1 and the productivity of the first market occupation (management), A_1, to be 1. Finally, as defined in Section 2, we set immigrant-specific preference shifters and compensation wedges to zero for natives: $\gamma^j_{1g} = 0$ and $\kappa^j_{1g} = 0$ for all g, j.

We use the remaining parameters to target the share of individuals (i, g) in j $\forall i, g, j$, the average annual earnings of individuals (i, g) in j relative to the average annual earnings of the base (type, subtype) (b, m) in $j = 1$, and the average hourly wages of individuals (i, g) in j relative to the average hourly wages of the base (type, subtype) (b, m) in $j = 1$.[8] In our analysis, we set the base (type, subtype) to be native males age 35 to 44 with a college degree.

3.3 Identification

Given the predetermined parameters, normalizations, and target moments, we back out the remaining parameters directly from the data. Our goal is to show how the model lets us recover the immigrant-specific barriers from observed data on occupational choices, earnings, and

[8] As shown in Table 3, we have more moments for annual earnings than hourly wages since wages are identical for all individuals in the non-market occupation given the linear production technology in this occupation. We set the hourly wage in this occupation in the model to be λ times the weighted average of wages across all market occupations in the data. Similarly, for each (type, subtype) pair, we set annual earnings in the non-market occupation in the model to be a λ of the weighted average of earnings across all market occupations in the data. In particular, we set $\lambda = 0.50$, which falls within the range of estimated replacement rates provided by unemployment insurance in the U.S. Appendix E provides our main results under alternative values of λ.

wages, and to investigate the features of the data that pin down each parameter. For analytical tractability, we focus on the case of perfect substitution across individuals in the inner nest: $\widetilde{\sigma}_j = \infty \; \forall j = 1, ..., J$. Appendix C.1 provides derivations of the equations used in this section.

Recovering the immigrant barriers requires identifying them from all other factors that make immigrants' outcomes differ from those of natives. The central challenge is to separately identify preferences ν_g^j and γ_{ig}^j, productivities z_{ig}^j, and compensation wedges τ_g^j and κ_{ig}^j. Building on Hsieh et al. (2019), we exploit the closed-form expressions that the Roy model with i.i.d. Fréchet draws delivers for sorting, earnings, and wages. These expressions let us recover the parameters sequentially, in three main blocks, where each step conditions on the objects already recovered and pins down one parameter from a distinct source of variation in the data. We set the population masses N_{ig} directly to their observed shares and proceed as follows.

First, we recover preferences. A key property of the model is that average earnings in an occupation are invariant to productivity and wedges: individuals with higher productivity draws sort into less desirable occupations and exactly offset them, so the cross-occupation pattern of average earnings within a group reflects differences in preferences. Thus, the common shifters ν_g^j follow from how natives' earnings vary across occupations, and the immigrant-specific shifters γ_{ig}^j follow from how immigrants' earnings depart from those of natives of the same subtype.

Second, given preferences, we recover productivities. A group's productivity in the non-market occupation z_{ig}^0 is inferred from how many of its members stay out of work and how much they earn when they work, relative to the base group. A group whose members remain non-employed even when their earnings are high must be very productive at home. The market productivities z_{ig}^j then follow from two quantities for each j, both measured relative to the non-market occupation: the share of the group that works in j, and the effective labor it supplies there, measured by earnings relative to hourly wages. A group is estimated to be more productive in an occupation the more its members sort into it and the more hours they supply once there.

Third, given preferences and productivities, we recover the compensation wedges, which come in two layers. The common wedge τ_g^j applies to natives and immigrants of a given subtype alike; it is inferred from how natives of that subtype are paid relative to the base subtype within an occupation, so that natives paid less than the base subtype in j are estimated to face a positive common wedge. The immigrant-specific wedge κ_{ig}^j, the barrier we ultimately seek to recover, is then identified from the hourly-wage gap between an immigrant group and its native counterpart within the same occupation. Since preferences and productivities have already been recovered from how groups sort and how much they work, this step isolates the part of that wage gap which those fundamentals, together with the groups' relative scarcity under imperfect substitution, cannot explain, and attributes it to the wedge. In the final step, we recover the occupation productivities A_j from the base type's relative labor supply across occupations. Because each parameter is pinned down by its own source of variation conditional

on the objects already recovered, the full set is identified sequentially and in closed form.

Using this strategy, the full set of parameters is identified from the three sets of moments documented in Figure 1 and Table 2: the occupational distribution, annual earnings, and hourly wages. Each parameter maps to the moment that disciplines it: preferences to the cross-occupation earnings pattern, productivities to earnings-to-wage ratios and employment shares, and compensation wedges to within-occupation wage gaps. Thus, we estimate the model with rich heterogeneity by linking the theoretical objects to observable variation in the data.

Population mass. In the model, the shares of individuals of each (i, g) pair are exogenous, so we set N_{ig} equal to their observed fraction in the data:

$$N_{ig} = \text{Fraction of individuals of type and subtype } (i, g).$$

Common and immigrant-specific preferences. The solution of the model implies:

$$\frac{\text{Earnings}_{ig}^{j}}{\text{Earnings}_{ig}^{k}} = \frac{\nu_g^k}{\nu_g^j} \times \frac{1+\gamma_{ig}^k}{1+\gamma_{ig}^j}, \tag{2}$$

where Earnings_{ig}^{j} is the geometric average annual earnings across all individuals of (i, g) in occupation j. In the model, the earnings are given by the right-hand side of the budget constraint.

Given that immigrant preference shifters are zero for natives (i.e., $\gamma_{1g}^j = 0 \; \forall g, j$) and that the preference for the non-market occupation is normalized to 1 (i.e., $\nu_g^0 = 1 \; \forall g$), writing Equation (2) for natives $i = 1$, for occupation j, and setting $k = 0$, we obtain:

$$\nu_g^j = \lambda \left(\frac{\text{Earnings}_{1g}^{j}}{\text{Avg. market earnings}_{1g}} \right)^{-1},$$

where $\text{Avg. market earnings}_{ig}$ denotes the weighted average of Earnings_{ig}^{j} across market occupations j, with weights given by the share of individuals of such type and subtype that choose each market occupation.[9] That is, the earnings of natives of subtype g in occupation j relative to their weighted average earnings across all occupations is informative about their preference for j. Using data on natives' earnings in each j for each g and data on natives' average market earnings for each g, this relationship allows us to obtain common preferences $\nu_g^j \; \forall g, j$.

Given common preferences $\{\nu_g^j\}_{g,j}$ and our normalization that immigrant preference shifters are zero in the non-market occupation (i.e., $\gamma_{ig}^0 = 0 \; \forall i, g$), we use Equation (2) to back out these preference shifters for every immigrant type and subtype (i, g) in market occupation j as follows:

$$1 + \gamma_{ig}^j = \lambda \left(\nu_g^j \frac{\text{Earnings}_{ig}^{j}}{\text{Avg. market earnings}_{ig}} \right)^{-1} = \left[\left(\frac{\text{Earnings}_{ig}^{j}}{\text{Earnings}_{1g}^{j}} \right) \Big/ \left(\frac{\text{Avg. market earnings}_{ig}}{\text{Avg. market earnings}_{1g}} \right) \right]^{-1}. \tag{3}$$

[9] Recall from Section 3.2 that, for each (type, subtype) pair, we set the annual earnings in the non-market occupation to be a fraction λ of the weighted average of annual earnings across all market occupations. This implies that $\text{Earnings}_{ig}^{0} = \lambda \times \text{Avg. market earnings}_{ig} \; \forall i, g$.

Immigrant preferences are identified by comparing the earnings of immigrants of type (i, g) in occupation j relative to their average earnings across occupations vis-a-vis the earnings of natives of subtype g in occupation j relative to their average earnings. Thus, given data on the earnings of immigrants and natives for each (type, subtype) pair for each occupation and their average earnings across occupations, we back out immigrant preference shifters $\gamma_{ig}^{j}\ \forall i, g, j$.

For instance, consider immigrants ($i > 1$) and natives ($i = 1$) of the same subtype g. Suppose that the average earnings across market occupations are higher for immigrants (i, g) than for natives $(1, g)$. Suppose further that the average earnings for immigrants are even larger than that for natives in j. In this case, the model attributes a lower preference for this occupation among immigrants (a lower γ_{ig}^{j}). That is, compared to a scenario where the earnings gap specific to j is equal to the average earnings gap, immigrants receive lower utility from working in j and thus need to be compensated with a larger positive earnings gap relative to natives in j.

Individual productivity: Non-market occupation. Consider individual (type, subtype) pair (i, g) in the non-market occupation. The solution of the model implies that:

$$z_{ig}^{0} = \left(\frac{\text{Avg. market earnings}_{ig}}{\text{Avg. market earnings}_{bm}}\right)^{\frac{1}{1+\xi}} \left(\frac{\text{Fraction of non-employed}_{ig}}{\text{Fraction of non-employed}_{bm}}\right)^{\frac{1}{\eta}}, \tag{4}$$

where (b, m) denotes the base group whose non-market productivity is normalized to one ($z_{bm}^{0} = 1$). Then, the productivity of (i, g) in the non-market occupation is identified from differences in average market earnings and the fraction of non-employed, relative to the base group. Thus, given data on the fraction of type-subtype pairs in the non-market occupation and their average market earnings, we obtain their productivities in the non-market occupation $z_{ig}^{0}\ \forall i, g$.

Consider individuals of type-subtype (i, g) and (b, m). First, assume for a moment that both groups have the same fraction of non-employed individuals. If the former group has higher average market earnings than the latter group, then it must be that the former group has higher productivity at the non-market occupation. Second, assume instead that both groups have the same average market earnings but the fraction of non-employed is higher in the former group. Then, it must be that the former group has higher productivity at the non-market occupation.

Individual productivity: Market occupations. Consider individual (type, subtype) pair (i, g) and some occupation j. The solution of the model implies that:

$$\frac{z_{ig}^{j}}{z_{ig}^{0}} = \frac{\text{Earnings}_{ig}^{j}/\text{Wages}_{ig}^{j}}{\text{Earnings}_{ig}^{0}/\text{Wages}_{ig}^{0}} \times \left(\frac{\text{Fraction of employed}_{ig}^{j}}{\text{Fraction of employed}_{ig}^{0}}\right)^{\frac{1}{\eta}}, \tag{5}$$

where $\text{Fraction of employed}_{ig}^{j}$ denotes the fraction of individuals of type (i, g) in occupation j, and Wages_{ig}^{j} is given by the geometric average of hourly wages across all individuals of type-subtype (i, g) in occupation j. In the model, the wage of an individual is given by $(1-\tau_{g}^{j}-\kappa_{ig}^{j})w_{ig}^{j}$. Then, we have that the productivities of individuals across market occupations are identified from

differences in the ratio of earnings to wages between market occupation j and the non-market occupation, as well as from the fraction of individuals employed in occupation j relative to the non-market occupation. As a result, we obtain their individual productivities $z_{ig}^{j}\ \forall i, g$ across market occupations. Equation (5) implies that individuals are estimated to be more productive in market occupation j relative to the non-market occupation if their earnings-to-wage ratio in j is higher or if a higher fraction of individuals chooses j than the non-market occupation.

This mapping of hourly wages also implies that annual hours are $\ell z_{ig}^{j}\varepsilon_j$, so that $z_{ig}^{j}\varepsilon_j$ reflects the ability to supply more hours, or the same hours at a lower disutility. We assess the role of this mapping through two alternatives in Section 6. First, we instead define the model's hourly wage as $w_{ig}^{j}z_{ig}^{j}\varepsilon_j$, where $z_{ig}^{j}\varepsilon_j$ reflects output per hour. In this case, the productivity and wedge parameters still remain analytically recoverable.[10] Second, a version with inelastic labor supply removes the need to target hourly wages, at the cost of recovering only occupation-independent productivities z_{ig}. Section 6 presents our main results under these two alternative assumptions.

Compensation wedges. We back out common compensation wedges $\{\tau_g^j\}$ by focusing on natives. Consider natives ($i = 1$) of subtypes g and m (base subtype). Then, we have that:

$$1 - \tau_g^j = \frac{\text{Wages}_{1g}^{j}}{\text{Wages}_{1m}^{j}}.$$

Thus, given data on wages of native (type, subtype) pairs across occupations, we obtain common compensation wedges $\tau_g^j\ \forall g, j$. This expression implies that the common compensation wedges in an occupation j are identified from data on the wages of natives relative to those of the base subtype for the given occupation. In particular, natives of subtype g whose wages in j relative to those of the base subtype are lower are inferred to have positive compensation wedges.

Next, we back out immigrant compensation wedges $\{\kappa_{ig}^j\}$. Let (i, g) denote an immigrant of a given type-subtype, and let $(1, g)$ denote her native counterpart. Then, the model implies:

$$\frac{1-\tau_g^j-\kappa_{ig}^j}{1-\tau_g^j} = \frac{\text{Wages}_{ig}^{j}}{\text{Wages}_{1g}^{j}} \left\{ \frac{\sum_{q\in\mathcal{I}_i}\sum_{r=1}^{G} N_{qr}\left(z_{qr}^{j}\right)^{1+\xi}\left[(1+\gamma_{qr}^{j})\nu_r^j\text{Wages}_{qr}^{j}\right]^{\xi}\left(p_{qr}^{j}\right)^{\frac{\eta-(1+\xi)}{\eta}}}{\sum_{r=1}^{G} N_{1r}\left(z_{1r}^{j}\right)^{1+\xi}\left[(1+\gamma_{1r}^{j})\nu_r^j\text{Wages}_{1r}^{j}\right]^{\xi}\left(p_{1r}^{j}\right)^{\frac{\eta-(1+\xi)}{\eta}}} \right\}^{\frac{1}{\sigma_j}}, \quad (6)$$

where $\mathcal{I}_i$ is the set of immigrant types. Then, using data on wages and allocations across (type, subtype) pairs, we obtain immigrant compensation wedges $\kappa_{ig}^j\ \forall i, g, j$.

This first term implies that immigrant compensation wedges are identified from information similar to that used for the common compensation wedges. Any under-compensation in wages relative to their native counterparts is interpreted as a positive immigrant compensation wedge. Critically, the wedge is identified separately from productivity. Productivity z_{ig}^{j} is recovered in an earlier step, from cross-occupational variation in a group's own employment shares and earnings-

[10] Recoverability requires that the wedges $(1 - \tau_g^j - \kappa_{ig}^j)$ and the productivity terms $z_{ig}^{j}\varepsilon_j$ enter the mapping non-multiplicatively; otherwise we could estimate only productivities independent of occupations, z_{ig}.

to-wage ratios (Equation 5), whereas the wedge is identified from within-occupation comparisons between an immigrant group and its native counterpart. Since productivity has already been recovered and is held fixed at this stage, the wedge does not simply absorb immigrant-native productivity differences; it captures the part of the immigrant-native wage gap that remains after accounting for productivity and, through the second term, relative scarcity.

The second term of the right-hand side arises from the imperfect substitutability between natives and immigrants. The numerator can be thought of as a measure of aggregate labor supply of all immigrants, which is proportional to population as well as occupation-specific productivity, hours worked, and allocations. The denominator is analogous for all natives. This term implies that differences in the relative supply between natives and immigrants are also captured by immigrant compensation wedges. For instance, if immigrants are a small fraction of the population but have similar productivities in occupation j, work similar hours, and are observed to be equally likely as natives to choose this occupation, then immigrant compensation wedges κ in this occupation would be positive. For κ to be zero, immigrants would need to be paid relatively more than natives given their relative scarcity.[11]

Occupation productivity. Consider the base type and subtype (b, m), along with two alternative market occupations j and k. Let k be given by the first occupation such that $A_k = 1$ given our normalizations. The solution of the model implies that:

$$A_j = \left\{ \left(\frac{\text{Wages}^j_{bm}}{\text{Wages}^1_{bm}} \right)^{\sigma} \frac{\sum_{g=1}^{G} N_{bg} \left(z^j_{bg}\right)^{1+\xi} \left[\nu^j_g \text{Wages}^j_{bg}\right]^{\xi} \left(p^j_{bg}\right)^{\frac{\eta-1}{\eta}}}{\sum_{g=1}^{G} N_{bg} \left(z^1_{bg}\right)^{1+\xi} \left[\nu^1_g \text{Wages}^1_{bg}\right]^{\xi} \left(p^1_{bg}\right)^{\frac{\eta-1}{\eta}}} \right\}^{\frac{1}{\sigma-1}}. \tag{7}$$

Note that all objects in this expression can be computed either directly from the data or indirectly using data along with the derivations above. Thus, Equation (7) allows us to obtain $A_j \; \forall j$. This expression contrasts the relative labor supply of the base type b between occupation j and the base occupation ($j = 1$). Controlling for differences in wages of the base (type, subtype) across occupations, if labor supply of the base type is higher in an occupation j relative to that in the base occupation, then occupation j is inferred to feature higher occupational productivity.

3.4 Properties of the efficient allocation

Before we estimate the immigrant wedges and study their macroeconomic effects, we investigate the model's equilibrium in the absence of immigrant-specific distortions. This efficient allocation serves as the benchmark against which all immigrant wedges are estimated.

[11] Immigrants from certain countries of origin may benefit from larger or more established communities, potentially allowing them to face lower labor market barriers. While we cannot identify such network effects in our data, our approach already implicitly captures these effects. In particular, Equation (6) implies that, conditional on other observables such as wages, hours worked, and masses of worker groups across occupations, if we observe a higher fraction of employment of certain (type, subtype) (i, g) in an occupation j relative to natives, then immigrant compensation wedges κ should be smaller for this immigrant group (i, g) in that occupation j.

To simplify our discussion, we focus on the case where natives and immigrants are perfect substitutes, i.e., $\sigma_j = \infty \, \forall j$.[12] We then define the efficient equilibrium allocation to be the equilibrium in which immigrant compensation wedges are set to zero, i.e., $\kappa_{ig}^j = 0 \; \forall i, g, j$. Thus, in this case, natives and all immigrant types are subject to the same level of distortions across occupations. We characterize a key property of the efficient allocation.

Equation (6) implies that natives and immigrants of any given type i with the same subtype g should earn the same average hourly wages in all occupations j. Employment shares and hours can still differ across these groups, since only the wedge is removed. Under the *i.i.d.* Frechet structure we adopt from McFadden (1972), Eaton and Kortum (2002), and Hsieh et al. (2019), productivity differences show up in where workers sort rather than in average earnings within an occupation. A more productive immigrant group, for instance, would sort more heavily into an occupation but still earn the same average hourly wage as natives there.

Is this property of the efficient allocation in line with the data? Results in Figure 1 and Table 2 show that immigrant-native average hourly wage gaps are nonzero across all occupations, and that the magnitude of these gaps differs substantially across occupations.[13] For example, relative to recent immigrants, the average hourly wage of natives is 48 percent (1.17/0.79) larger in routine manual occupations and 5 percent (1.85/1.94) lower in non-routine cognitive occupations.[14] Thus, our empirical findings reject the predictions of the efficient allocation.

4 Immigrant Wedges

We now study the extent and implications of immigrant wedges. Section 4.1 estimates the model parameters and wedges using the approach described above. Section 4.2 provides evidence validating our interpretation of the model-implied wedges as meaningful immigrant barriers.

4.1 Estimates of immigrant wedges and productivities

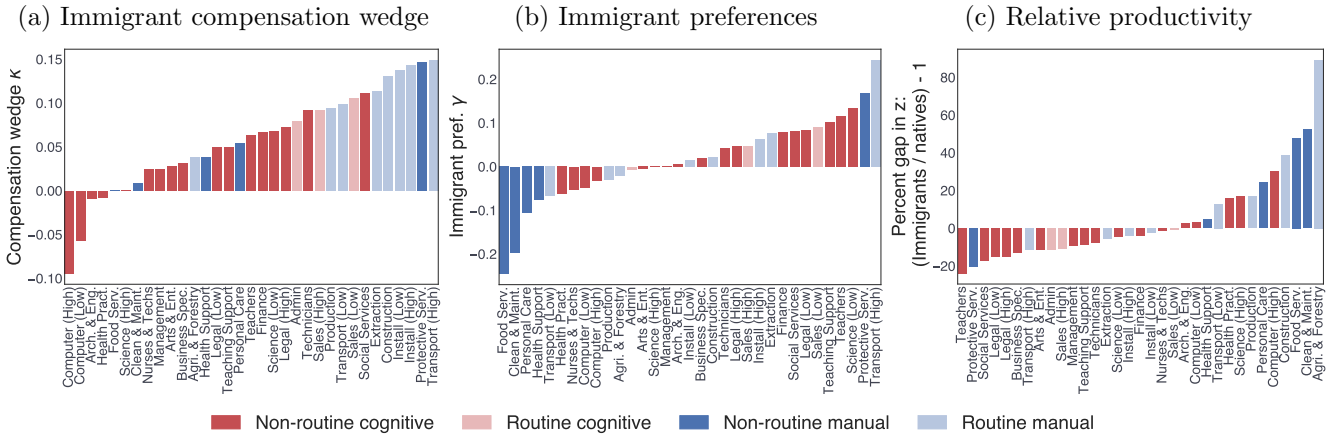
We begin by presenting our estimates of immigrant wedges (κ), preferences (γ), and productivity (z). Figure 2 presents averages across market occupations, and Table 4 presents averages across immigrant types. Heterogeneity in the estimated wedges can shed light on the mechanisms underlying them, while also serving to externally validate the reasonability of our estimates. Given the large number of parameters of our model (31,510 parameters, as described

[12] Our calibration sets $\sigma_j = 20 \, \forall j$, motivated by empirical studies on the degree of substitutability between immigrant and native labor supply, and we use perfect substitution only for this comparison. Even without perfect substitution, we derive the key equations analytically (see Section 3.3), so the more general case remains tractable. Moreover, $\sigma_j = 20$ implies a high degree of substitutability, close to the perfect substitution case. For these reasons, the discrepancy between our economy and the efficient allocation benchmark is second order.

[13] To be clear, results in Figure 1 do not condition on a specific immigrant type i and subtype g, and results in Table 2 only condition on immigrant type but do not condition on subtype g. However, even when we focus on a specific immigrant type i and compare outcomes in the data between immigrant type i and natives with the same subtype g, this property of the efficient allocation is rejected by the data.

[14] These are largely unchanged when conditioning on observables and comparing outcomes within an occupation.

Figure 2: Immigrant compensation wedges, labor supply preferences, and relative productivity



Notes: This figure plots the average values of immigrant compensation wedges κ, immigrant preferences γ, and the percent gap in productivity z between immigrants and natives (calculated as immigrants/natives -1) across market occupations. The legend classifies occupations into non-routine cognitive, routine cognitive, non-routine manual, and routine manual groups.

in Table 3), we restrict attention to weighted averages of the estimated parameters wherever necessary. Table A3 shows that the model closely matches the distribution of individuals across occupations as well as their annual earnings and hourly wages in the data shown in Table 2.

Immigrant wedges and productivities across occupations. Figure 2 plots average immigrant compensation wedges κ, immigrant preference shifters γ, and the percent gap in productivity z between immigrants and natives, measured as immigrants/natives -1, across market occupations. As seen, compensation wedges are mostly positive, suggesting that immigrants face meaningful barriers in many occupations. These wedges tend to be largest in routine manual occupations such as transportation, installation, construction, and extraction, whereas non-routine cognitive occupations are ranked much lower. At the same time, some non-routine cognitive occupations, such as high- and low-skill computer and mathematical jobs, engineering, and health practitioner roles, exhibit negative immigrant compensation wedges.

The immigrant preference shifters are also dispersed across occupations. Intuitively, if an occupation appears relatively favorable for immigrants in terms of productivity and compensation, yet immigrants are still less likely to work in it than observationally similar natives, the model attributes part of this to a lower preference for that occupation, as captured by γ. The most negative values of γ are concentrated in non-routine manual occupations, particularly food service, cleaning and maintenance, personal care, and health support. By contrast, the most positive preference shifters are in transportation (high), protective services, and many non-routine cognitive jobs such as teaching and some legal (low) and science-related (low) occupations.

Finally, the productivity estimates imply substantial heterogeneity across occupations. Relative to natives, immigrants are more productive in many manual occupations, including agriculture and forestry, cleaning and maintenance, food service, construction, and personal care, and they are also more productive in selected non-routine cognitive occupations that are more

Table 4: Estimation results

	Immigrant compensation wedge κ								Common
Occupation type	N	$I_{0\text{-}4}$	$I_{5\text{-}10}$	I_{10+}	$I_{\text{Low Eng}}$	$I_{\text{High Eng}}$	I_{Undoc}	I_{Doc}	comp. wedge τ
Non-routine cognitive	0.00	0.06	0.02	0.01	0.12	-0.01	-0.01	0.03	0.23
Non-routine manual	0.00	0.09	0.06	0.00	0.04	0.00	0.04	0.01	0.31
Routine cognitive	0.00	0.16	0.12	0.07	0.14	0.06	0.11	0.08	0.39
Routine manual	0.00	0.18	0.14	0.09	0.13	0.06	0.14	0.08	0.22
Non-market	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	Immigrant preference γ								Common
Occupation type	N	$I_{0\text{-}4}$	$I_{5\text{-}10}$	I_{10+}	$I_{\text{Low Eng}}$	$I_{\text{High Eng}}$	I_{Undoc}	I_{Doc}	pref. ν
Non-routine cognitive	0.00	0.03	0.01	0.01	-0.09	0.04	0.04	0.00	0.46
Non-routine manual	0.00	-0.11	-0.19	-0.17	-0.23	-0.06	-0.21	-0.14	0.77
Routine cognitive	0.00	0.12	0.04	0.02	-0.04	0.07	0.01	0.04	0.57
Routine manual	0.00	0.04	-0.01	-0.03	-0.04	0.04	-0.01	-0.02	0.54
Non-market	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.00
	Worker productivity z								Occupation
Occupation type	N	$I_{0\text{-}4}$	$I_{5\text{-}10}$	I_{10+}	$I_{\text{Low Eng}}$	$I_{\text{High Eng}}$	I_{Undoc}	I_{Doc}	prod. A
Non-routine cognitive	0.61	0.57	0.58	0.57	0.54	0.58	0.58	0.56	0.82
Non-routine manual	0.53	0.63	0.76	0.70	0.77	0.59	0.80	0.63	0.42
Routine cognitive	0.72	0.55	0.63	0.67	0.65	0.65	0.64	0.66	0.68
Routine manual	0.68	0.75	0.86	0.84	0.89	0.70	0.91	0.76	0.50
Non-market	0.84	0.79	0.75	0.78	0.69	0.89	0.74	0.80	0.25

Notes: This table presents estimated common compensation wedges τ, immigrant compensation wedges κ, common preference shifter ν, immigrant preference shifter γ, individual productivity z, and occupation productivity A. For expositional purposes, we report these outcomes across four broad occupation categories, where we assign 33 market occupations into categories based on their skill and task-intensity: non-routine cognitive, non-routine manual, routine cognitive, and routine manual.

technical, such as high-skill computer and mathematical jobs and high-skill science occupations. However, immigrants appear less productive in teaching, protective services, social services, and legal occupations. Overall, the estimates suggest that immigrants have a relative productivity advantage in several manual, service, and technical occupations.

Taken together, our estimates reveal large differences in immigrant barriers, preferences, and productivities across occupations. For instance, immigrants are often more productive than natives in manual occupations, but they also face substantial barriers in these occupations. These estimates suggest that a model that accounts for immigrants' heterogeneous outcomes across occupations is critical for understanding the aggregate implications of immigrant barriers.

Immigrant wedges and productivities across immigrant types. We now examine the extent to which immigrant wedges and productivities differ across immigrant types. Table 4 reports weighted averages of immigrant compensation wedges κ_{ig}^{j}, preference shifters γ_{ig}^{j}, and productivities z_{ig}^{j}. While we focus on immigrant wedges and productivities, we also report common compensation wedges τ_{g}^{j} and preference shifters ν_{g}^{j}, and occupation productivities A_j.

We find large differences in immigrant barriers and productivities by time since arrival. Re-

cent immigrants face larger compensation wedges than intermediate and established immigrants for all occupation groups. Moreover, intermediate immigrants lie between recent and established immigrants in κ, consistent with gradual integration over time. Recent immigrants are also less productive, on average, in market occupations than immigrants with longer tenure, except in non-routine cognitive occupations. Relative to natives, immigrants tend to be more productive in manual occupations and less so in cognitive ones, especially among recent immigrants.

Differences by English fluency are even more pronounced. Non-fluent immigrants face substantially larger compensation wedges than fluent immigrants across all occupation groups. They also exhibit stronger distaste for non-routine jobs. Non-fluent immigrants are also considerably more productive in manual jobs than their fluent counterparts. This is consistent with the lower language demands typically associated with these types of jobs.

In terms of legal status, undocumented immigrants face larger barriers than documented immigrants; but this is not true for the small share who are in non-routine cognitive occupations. Undocumented immigrants are more productive in manual jobs than documented immigrants.

Broadly, these results show that the relationship between immigrant barriers and productivity differs sharply across immigrant types and occupations. In manual occupations, immigrants who face larger barriers often also display higher relative productivity, suggesting that the barriers captured by κ are not simply proxies for low productivity. In non-routine cognitive occupations, by contrast, non-fluent immigrants tend to face both higher wedges and lower relative productivity, consistent with these jobs relying more heavily on communication, local knowledge, and formal access. Finally, the immigrant-specific preference shifters γ are not monotone across subgroups in the way that κ generally is, consistent with our interpretation that γ captures preference heterogeneity rather than labor-market barriers. For example, while compensation wedges are higher for recent than for established immigrants across all occupation groups, the corresponding γ values do not move as uniformly and may even switch sign across occupations. These patterns suggest that compensation wedges and preference heterogeneity capture distinct margins in the data, with the systematic barrier-like variation reflected in κ rather than in γ.

4.2 Immigrant wedges: Validation exercises

A potential concern about the interpretation of our estimated wedges is that they might reflect productivity differences rather than true immigrant barriers. We now provide additional evidence to validate our interpretation of the wedges as meaningful barriers faced by immigrants.

Immigrant wedges by time since immigration. We first examine how our estimated immigrant compensation wedges evolve as immigrants spend more time in the host country. If the wedges partly reflect barriers that weaken with labor-market integration, rather than only *persistent* productivity differences, we would expect them to decline with time since arrival. This is indeed what we find in Table 4. Compensation wedges decline monotonically with time since

Table 5: Immigrant compensation wedges vs. external evidence

External measure	Immigrant groups	Correlation: External measure vs wedges
Fraction of jobs requiring a license	All immigrants	-0.05
	0–4 years	0.27
	5–10 years	-0.10
	>10 years	-0.11
	Documented	0.05
	Undocumented	0.58
Language intensity	All immigrants	-0.11
	English fluent	-0.08
	English non-fluent	0.46

Notes: This table reports correlations between external occupation-level measures and model-implied immigrant compensation wedges across occupations in the U.S. The first panel reports correlations between the fraction of jobs requiring a license and model-implied immigrant compensation wedges. We report the correlation for all immigrants, and then separately by tenure and legal status. We use CPS data between 2016 and 2019 to calculate the fraction of jobs requiring a license for each market occupation, using the same occupation definitions as in Section 3.1. The second panel reports correlations between an occupation-level language-intensity index and model-implied immigrant compensation wedges. The language-intensity index is the average of O*NET importance scores for reading comprehension, writing, speaking, and active listening, normalized to lie between 0 and 1. We aggregate the index to the 33 market occupations and correlate each occupation's index with the corresponding average immigrant compensation wedge.

arrival: they are largest for recent immigrants (0–4 years), smaller for those with intermediate tenure (5–10 years), and smallest for established immigrants (> 10 years) in all broad occupation groups. For example, in routine manual occupations, the average compensation wedge falls from 0.18 for recent immigrants to 0.14 for intermediate immigrants and 0.09 for established immigrants. Because the model allows occupation-specific productivity to differ across tenure groups, the decline in estimated wedges is measured *after* accounting for these productivity differences. It therefore supports the interpretation that the wedges partly capture barriers that weaken as immigrants adjust to the U.S. labor market, rather than solely persistent native–immigrant productivity gaps.[15] This pattern is also consistent with previous work emphasizing a period of adjustment for newly arrived immigrants (e.g., Dostie, Li, Card, and Parent 2020).

While Table 4's estimates are internally consistent with this interpretation, it is useful to validate them against observable barriers measured externally from the data. We therefore compare the model-implied immigrant compensation wedges with occupational licensing requirements, a prominent barrier faced by newcomers. Since 2016, the Current Population Survey (CPS) reports whether a respondent's current job requires a government-issued professional, state, or industry license. We pool the CPS data between 2016 and 2019 and calculate the fraction of jobs requiring a license for each market occupation.[16] As expected, the fraction of jobs requiring a license is highest in healthcare, legal, education, healthcare support, and protective services occupations, and lowest in cleaning and maintenance, administrative support, and agriculture.

The top panel of Table 5 reports correlations between these licensing shares and the model-implied immigrant compensation wedges across occupations. For immigrants overall, the corre-

[15] Unobserved differences across arrival cohorts may also contribute to this result.

[16] Using the CPS, we also apply the same sample selection and definition of employment as in Section 3.1.

lation is small and slightly negative. However, when split by time since arrival, the correlation is positive and large for recent immigrants and turns modestly negative for intermediate and established immigrants. This suggests that licensing requirements are especially important barriers for new immigrants, but they weaken over time as immigrants acquire credentials, information, or other occupation-specific requirements needed to work in licensed jobs. More broadly, that the relationship between wedges and licensing is strongest for recent immigrants and fades with tenure provides an external validation for our estimates. If wedges mainly captured persistent unobserved productivity differences between natives and immigrants, rather than barriers, there would be little reason to expect this relationship to weaken systematically with tenure.

Immigrant wedges by legal status. Compensation wedges are also larger for undocumented immigrants, supporting the reasonableness of our estimates. Table 4 shows that undocumented immigrants face larger compensation wedges than documented immigrants, consistent with the idea that they encounter greater labor-market barriers. Table 5 provides external support for this interpretation. The strong positive correlation between wedges and licensing requirements for undocumented immigrants suggests that licensing barriers are especially binding for this group. For documented immigrants, the correlation is positive but close to zero. This suggests that, while licensing requirements act as barriers, documented immigrants are more likely to be able to acquire credentials or otherwise overcome them. Undocumented immigrants, by contrast, lack the legal status needed to enroll in educational programs, apprenticeships, or licensing processes.

Immigrant wedges and language barriers. Finally, we compare the model-implied compensation wedges with an occupation-level measure of language intensity. If wedges partly capture barriers to effective communication in the labor market, then they should be more strongly related to language-intensive occupations for immigrants who are not fluent in English. To evaluate this hypothesis, we construct a language-intensity measure using O*NET. Specifically, we average occupation-level importance scores for reading comprehension, writing, speaking, and active listening, normalize the resulting index to lie between 0 and 1, and aggregate it to the same 33 market occupations used in the analysis. The bottom panel of Table 5 shows that for the average immigrant, the correlation between an occupation's language intensity and the model-implied compensation wedge is small and negative. The same is true for English-fluent immigrants. In contrast, the correlation is strongly positive for non-English-fluent immigrants. This pattern is consistent with the view that language-related barriers are larger among immigrants for whom English fluency is likely to be most limiting. It also addresses the concern that the estimated wedges simply reflect broad occupation-level productivity differences. If that were the case, then there would be little reason for the relationship with language intensity to appear primarily for non-English-fluent immigrants and not for English-fluent immigrants.

Alternative modeling assumptions. In addition to the results above, we also our assess our modeling assumptions on immigrant productivity. In Section 6, we relax our baseline assumption of identical underlying productivity distributions for immigrants and natives and explore alternative mappings between productivity differences and observable outcomes in the data.

5 Immigrant Wedges: Policy Implications

The following section analyzes the macroeconomic effects of policies that reduce immigrant wedges, both in the aggregate (Section 5.1) and across the distribution (Section 5.2).

5.1 Aggregate implications of immigrant wedges

In this section, we conduct counterfactuals that reduce immigrant barriers. As a benchmark, we begin with a counterfactual in which all immigrant compensation wedges are removed, meaning that immigrants now face the same common compensation wedges as natives. We then turn to more targeted counterfactuals that are closer to implementable policy interventions than fully eliminating wedges. We implement four main exercises. The first sets the compensation wedges of undocumented immigrants equal to those of documented immigrants, meant to resemble an amnesty. The second sets the compensation wedges of non-fluent immigrants equal to those of fluent immigrants, meant to resemble language training, social integration, and the enhancement of workplace-relevant communication skills. The third sets the compensation wedges of recent immigrants equal to those of intermediate-tenure immigrants, meant to resemble policies that enable integration into the labor market. The final counterfactual simultaneously equalizes three dimensions of immigrant compensation wedges: legal status, English-language fluency, and tenure. We implement these experiments by assigning undocumented immigrants the wedges of their documented counterparts, non-fluent immigrants the wedges of their fluent counterparts, and recent immigrants the wedges of their intermediate-tenure counterparts. Importantly, unlike the counterfactual where all immigrant wedges are removed, the remaining counterfactuals only set one subgroup's wedges equal to another's, without removing all immigrant barriers entirely.

Aggregate real GDP gains. Table 6 presents the aggregate effects of each counterfactual on real GDP, TFP, employment, hours, and the immigrant share. The full removal of immigrant compensation wedges raises real GDP by 1.54 percent and, while arguably infeasible, provides a natural benchmark for comparison. The gains under this scenario are driven primarily by higher employment, while TFP and especially hours gains are relatively smaller.

The more targeted counterfactuals generate smaller and more varied gains. For example, the legal-status and tenure counterfactuals yield little to no output gain despite increasing employment, because these gains are offset by a similar decline in hours. By contrast, the language counterfactual raises real GDP by 0.41 percent, through higher employment and TFP gains.

The counterfactual that simultaneously reduces legal-status, language, and tenure-related

Table 6: Aggregate effects across counterfactuals

	Percent change				Change in
Counterfactual	Real GDP	TFP	Employment	Hours	immigrant share (pp)
No compensation wedges	1.54	0.21	1.31	0.02	1.12
Undocumented → documented	-0.02	-0.20	0.38	-0.20	0.32
Non-fluent English → fluent English	0.41	0.20	1.00	-0.78	0.85
Recent (0–4) → intermediate (5–10)	0.04	0.09	0.26	-0.31	0.23
Combined legal, language, and tenure	0.83	-0.00	1.31	-0.46	1.10

Notes: This table presents the aggregate effects of each counterfactual on real GDP, TFP, employment, hours, and the immigrant share. Real GDP, TFP, employment, and hours are reported as percent changes relative to the baseline. The change in immigrant share is reported in percentage points (pp). Aggregate real GDP is output produced in the market sector, TFP is real GDP per hour, employment is the mass of workers in market occupations, and hours is average hours worked in market occupations. The final row reports the counterfactual that jointly reduces legal-status, English-proficiency, and tenure-related barriers.

barriers increases real GDP by 0.83 percent, accounting for more than half of the gains from fully removing immigrant-specific compensation wedges. The fact that the combined counterfactual's gains are larger than the sum of the individual targeted counterfactuals is evidence of the quantitative importance of interactions and complementarities across barriers. Importantly, it also offers an explanation for why reducing barriers along a single dimension may not be as effective. This result is also consistent with the subgroup correlations found in the data. For example, undocumented immigrants are more likely to be non-fluent and recent, so policies that reduce barriers along only one dimension naturally leave many correlated barriers untouched. In practice, this points to the value of a multidimensional approach to reducing immigrant barriers. For example, legalizing an undocumented immigrant may be less effective if other barriers, such as weak language skills or non-transferability of credentials, hinder entry into suitable jobs.

It is also useful to consider the share of immigrants directly affected by each policy. In our data, undocumented, non-fluent, and recent immigrants account for about 36, 50, and 13 percent of immigrants, respectively. These differences in group size provide context for the aggregate gains, but they are not sufficient on their own to explain the quantitative differences across counterfactuals. For example, despite both groups representing large shares of immigrants and differing only modestly in size, the legal-status counterfactual generates essentially no aggregate output gain, whereas the language counterfactual raises real GDP by 0.41 percent. Likewise, although recent immigrants account for a non-trivial share of immigrants, the tenure counterfactual delivers gains that are roughly ten times smaller than those under the language counterfactual. Overall, the size of the exposed subgroup alone cannot explain the aggregate effects. The gains also depend on the magnitude and type of wedges being reduced, the occupations toward which workers are reallocated, and how strongly the relevant barriers distort labor-market allocations.

5.2 Sources and heterogeneity of real GDP gains

Having summarized the aggregate effects, we now examine the mechanisms through which each counterfactual operates and how its effects vary across occupations and workers. For each

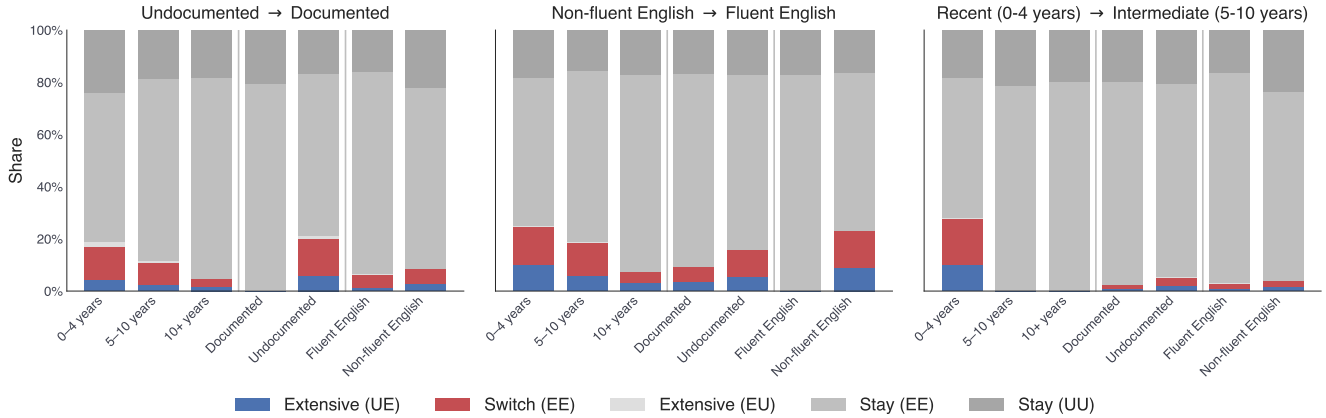
Table 7: Occupation-level effects across counterfactuals

	Percent change				Change in
Occupation type	Real GDP	TFP	Employment	Hours	immigrant share (pp)
Undocumented → Documented					
Aggregate	-0.02	-0.20	0.38	-0.20	0.32
Non-routine cognitive	-0.42	-0.25	0.46	-0.63	0.40
Non-routine manual	0.30	0.09	-0.15	0.36	-0.09
Routine cognitive	-0.27	-0.06	-0.35	0.14	-0.28
Routine manual	1.31	0.11	1.37	-0.17	1.02
Non-market	-0.57	0.03	-1.35	0.77	-1.18
Non-fluent English → Fluent English					
Aggregate	0.41	0.20	1.00	-0.78	0.85
Non-routine cognitive	0.30	0.15	1.89	-1.71	1.61
Non-routine manual	-0.59	0.12	-1.26	0.57	-0.87
Routine cognitive	0.46	0.26	0.77	-0.57	0.72
Routine manual	1.14	0.27	1.23	-0.36	0.96
Non-market	-2.21	0.13	-3.55	1.25	-3.18
Recent (0–4 years) → Intermediate (5–10 years)					
Aggregate	0.04	0.09	0.26	-0.31	0.23
Non-routine cognitive	0.28	0.08	0.82	-0.62	0.70
Non-routine manual	-0.18	0.05	-0.14	-0.09	-0.08
Routine cognitive	-0.21	0.01	-0.08	-0.14	-0.05
Routine manual	-0.22	0.01	-0.12	-0.11	-0.07
Non-market	-0.68	0.04	-0.93	0.22	-0.82
Combined legal, language, and tenure adjustment					
Aggregate	0.83	-0.00	1.31	-0.46	1.10
Non-routine cognitive	0.17	0.03	1.38	-1.22	1.20
Non-routine manual	0.60	0.10	-0.59	1.10	-0.37
Routine cognitive	0.56	0.26	0.62	-0.31	0.60
Routine manual	3.24	0.39	3.30	-0.45	2.45
Non-market	-2.53	0.17	-4.70	2.11	-4.25

Notes: This table presents the aggregate and occupation-level effects of each counterfactual on real GDP, TFP, employment, hours, and the immigrant share. The first four columns report percent changes relative to the baseline, while the final column reports percentage-point changes in the immigrant share. Aggregate real GDP is output produced in the market sector, TFP is real GDP per hour, employment is the mass of workers in market occupations, and hours is average hours worked in market occupations. Occupation rows report the corresponding changes within each broad occupation group. For expositional purposes, we group the 33 market occupations into four broad categories based on skill and task intensity: non-routine cognitive, non-routine manual, routine cognitive, and routine manual. Non-market refers to the home sector (i.e., non-employment).

counterfactual, we complement the aggregate effects in Table 6 with four sets of results. First, we decompose output gains by broad occupation groups and by the 33 detailed market occupations to identify where gains and losses arise. Second, we isolate the role of movements into and out of market work by comparing the full counterfactual with a version that allows only within-market reallocation. Third, we decompose individual transitions into movements from non-employment to employment, movements from employment to non-employment, switches across market occupations, and stayers. Fourth, we compare the responses of immigrants and natives

Figure 3: Decomposition of transition responses by immigrant subgroup



Notes: This figure plots the decomposition of transition responses across immigrant subgroups for three counterfactuals: undocumented → documented, non-fluent English → fluent English, and recent (0–4 years) → intermediate (5–10 years). Within each panel, bars report the share of the total transition response accounted for by five components: extensive margin from non-employment to employment (UE), market-to-market occupation switching (EE switch), extensive margin from employment to non-employment (EU), stayers in employment and the same occupation (EE stay), and stayers in non-employment (UU). Each bar sums to one. Subgroups are organized by tenure, legal status, and English proficiency. The blue and red segments highlight the UE and EE switch components, while the gray segments denote the remaining adjustment channels.

Table 8: Immigrant and native responses across counterfactuals

	Undocumented → Documented				Non-fluent → Fluent			
	Emp.	Hours	Wages	Earn.	Emp.	Hours	Wages	Earn.
Native aggregate	-0.02	-0.07	-0.02	-0.10	-0.06	-0.78	-0.07	-0.85
Immigrant aggregate	2.05	-0.70	-0.65	-1.04	5.44	-0.65	1.56	7.12
Non-routine cognitive	2.84	-3.54	-1.75	-5.88	12.44	-6.05	-0.79	-0.23
Non-routine manual	-0.31	1.48	0.87	4.75	-4.40	4.29	0.73	10.81
Routine cognitive	-2.15	1.36	-0.40	1.81	5.96	0.90	0.39	8.69
Routine manual	5.45	-0.33	0.26	5.25	5.19	0.89	0.79	11.91
Non-market	-7.61	4.60	0.00	4.60	-20.24	11.79	0.00	11.79
	Recent (0–4) → Intermediate (5–10)				Combined legal, language, and tenure			
	Emp.	Hours	Wages	Earn.	Emp.	Hours	Wages	Earn.
Native aggregate	-0.02	-0.20	-0.02	-0.23	-0.08	-0.68	-0.09	-0.77
Immigrant aggregate	1.45	-0.71	0.67	1.62	7.13	0.51	0.90	6.63
Non-routine cognitive	5.34	-2.69	-0.05	-0.10	9.15	-3.81	-0.73	-1.61
Non-routine manual	-0.53	0.25	0.23	1.07	-2.14	6.05	1.15	16.40
Routine cognitive	-0.36	0.23	-0.07	0.76	5.10	2.00	0.24	9.49
Routine manual	-0.27	0.09	-0.00	0.88	13.49	0.31	1.90	16.45
Non-market	-5.41	1.99	0.00	1.99	-26.51	17.77	0.00	17.77

Notes: This table presents the responses of immigrants and natives to each counterfactual. Responses are percent changes relative to the baseline. In each panel, the first row reports the aggregate response for natives, the second row reports the aggregate response for immigrants, and the remaining rows report immigrant responses by broad occupation group. Employment is the mass of workers in market occupations, hours is average hours worked in market occupations, wages is average hourly wages, and earnings is average annual earnings. For expositional purposes, we group the 33 market occupations into four broad categories based on skill and task intensity: non-routine cognitive, non-routine manual, routine cognitive, and routine manual. Non-market refers to the home sector.

in terms of employment, hours, wages, and earnings.

Legal-status counterfactual. We begin with the counterfactual that sets the compensation wedges of undocumented immigrants equal to those of documented immigrants. Table 7 reports the decomposition of real GDP, TFP, employment, and hours across broad occupation groups, while Figure A2 in Appendix D presents output gains and their sources across all 33 occupations. Together, they show that the legal-status counterfactual generates negligible changes in real GDP: output rises in manual jobs (especially routine ones), but this is offset by a decline in output in cognitive jobs. In terms of employment, routine manual jobs experience the largest increase. While non-routine cognitive jobs also see a rise in employment, declines in TFP and especially in hours dominate, leading to output losses. The decline in TFP and hours in these jobs implies that the rise in employment is concentrated among lower-productivity immigrants.

Table A4 reveals more about reallocation patterns by shutting down movements between non-employment and market (UE) occupations. When UE movements are prevented, employment gains in routine manual jobs fall the most, while cognitive jobs see a more modest difference. This indicates a two-tiered reallocation where movements out of non-employment feed into manual jobs, while within-market reallocation sees incumbents move to non-routine cognitive jobs.[17]

Figures 3 and A3 decompose these responses across immigrant and demographic groups. Figure 3 shows that both market-market (EE) and non-market to market (UE) transitions are mostly attributable to the undocumented, who are also predominantly recent and less fluent. However, within-market transitions are more prevalent. Figure A3 shows that transitions (UE or EE) are slightly more likely among the young, men, and those without a high school degree.

Finally, Table 8 compares the responses of immigrants and natives *separately*. It shows that immigrants experience larger employment and hours changes. Non-routine cognitive occupations observe large employment gains, but average hours, wages, and earnings fall because negative selection into those occupations, as lower-productivity workers enter. Routine manual occupations also see large employment gains, and average wages and earnings rise. Importantly, the effects on natives are as expected in sign but quantitatively negligible. Natives observe only very small declines in employment, hours, and wages, a pattern broadly consistent with the wider empirical literature finding that immigration has little effect on native-born workers' employment overall and only small average effects on wages (Card, 1990; Peri and Yasenov, 2019).[18]

Language counterfactual. Table 7 shows that the language counterfactual results in a modest but positive real GDP gain of 0.41 percent. The higher gains are due in part to a larger share of non-fluent immigrants compared to undocumented immigrants (50 percent versus 36 percent).

[17]This two-tiered reallocation pattern is reminiscent of Mandelman and Zlate (2022), who show that low-skilled immigration raises employment in low-skill jobs while putting downward pressure on wages. In our setting, removing immigrant wedges likewise draws workers from non-employment into manual occupations, though through improved immigrant integration rather than an immigration inflow.

[18]While the effects on natives overall are negligible, we find that counterfactuals targeting low-skilled or low-educated workers affect low-educated natives somewhat more than the average native, consistent with the degree of substitutability between the two groups. Nonetheless, the effects remain small.

Output and employment rise for all broad occupation groups except non-routine manual occupations, with the largest gains occurring in routine manual occupations, as depicted in Figure A2. TFP also experiences modest increases for all broad occupation groups, while hours decline in all except non-routine manual occupations. Table A4 shows that, when only within-market reallocation is allowed, employment changes in cognitive jobs are similar to those in Table 7. By contrast, without flows from the non-market occupation, the employment response in manual jobs declines the most. Taken together, this again points to two-tiered reallocation, as in the previous counterfactual. Figures 3 and A3 show that this reallocation is again concentrated among certain groups: market-market (EE) and non-market to market (UE) switches are more likely to occur for non-fluent and recent immigrants, as well as among the old, women, and those without a high school degree. Table 8 shows that immigrants experience larger changes in employment, wages, and earnings, while natives observe only a very small decline in these outcomes.

Overall, the language counterfactual appears quantitatively more important than the legal-status counterfactual, partly because it affects a broader group of immigrants, but the underlying mechanism remains similar: movements out of non-employment continue to matter especially for manual jobs, while within-market transitions reallocate workers to cognitive jobs.

Tenure counterfactual. Table 7 shows that the tenure counterfactual generates negligible output gains of 0.04 percent. Output in non-routine cognitive jobs rises, but this is offset by declines in other occupation groups. Employment follows similar patterns, while hours declines in all occupation groups. What is distinct about this counterfactual relative to the previous two is that the increases in employment and output are much more concentrated in non-routine cognitive occupations. This is seen clearly when ranking occupations by output gains in Figure A2. Importantly, among the counterfactuals considered so far, this one generates the least movement out of the non-market occupation. Table A4 reinforces this point: unlike the previous two counterfactuals, restricting adjustment to within-market reallocation yields movements that are not very different from those in the case where all flows are allowed. This suggests that integrating recent immigrants has broader effects within the market sector, rather than primarily pulling non-employed workers into manual jobs, as was the case in the earlier counterfactuals.

Figure A3 also shows that, unlike the previous counterfactuals, the college-educated account for relatively more transitions. Overall, the tenure counterfactual is distinct from the others in that it diverts more highly educated workers toward non-routine cognitive jobs rather than drawing non-market workers into manual jobs. Table 8 again shows negligible effects for natives, indicating that the main responses remain concentrated among immigrants.

Joint counterfactual. That the output and employment effects of the previous three counterfactuals are relatively modest is unsurprising. A key feature of the data is that many immigrant subgroups are correlated with one another. For example, undocumented immigrants are more

likely to be non-fluent, recent, less educated, and young. As a result, lowering barriers along one dimension may not lead to significantly larger gains if associated barriers remain in place. For example, amnesty may not be so effective if immigrants still face strong language and labor-market barriers. This is precisely why the joint counterfactual is useful as it allows us to assess how much larger the gains can be when multiple correlated barriers are reduced simultaneously.

As Table 7 shows, reducing legal-status, language, and tenure-related barriers jointly raises output across occupations, with all broad occupation groups experiencing an increase in either employment or hours. At the same time, Table A4 shows that the two-tiered reallocation pattern remains present in this setting. When movements out of the home sector are shut down, manual jobs experience the largest declines in employment relative to the full counterfactual that allows extensive-margin reallocation. This indicates that movements out of non-employment continue to feed disproportionately into manual jobs, while within-market reallocation remains important for non-routine cognitive occupations. Overall, the joint counterfactual reinforces the earlier message from the aggregate results: because the barriers faced by immigrants are correlated, reducing them jointly is substantially more effective than relaxing any single margin in isolation.

5.3 Welfare effects

Welfare calculation. Having examined the aggregate and heterogeneous labor-market effects of the different counterfactuals, we now turn to their welfare implications. In particular, we ask how the changes in labor market outcomes associated with each counterfactual translate into welfare gains or losses for the overall economy and for different groups of workers.

We evaluate welfare using a compensating-variation measure. Plugging the budget constraint into the utility function in Equation (1), utility for type-subtype (i, g) in occupation j is

$$U_{ig}^{j} = (1 + \gamma_{ig}^{j})\nu_{g}^{j} \left[\frac{1}{p}(1 - \tau_{g}^{j} - \kappa_{ig}^{j})\, w_{ig}^{j} z_{ig}^{j} \varepsilon_{j}^{\mu} (1 + s)\, \ell \right] - \frac{\ell^{\,1+\frac{1}{\xi}}}{1 + \frac{1}{\xi}}.$$

Now, consider two economies: baseline (B) and counterfactual (CF). They may differ in their immigrant compensation wedges κ_{ig}^{j} and in the rebate s, which in turn may affect equilibrium wages, occupation choices, and hours. In our simulations, we hold fixed each individual's vector of idiosyncratic occupation draws, $\{\varepsilon_j\}_{j=0}^{J}$, across the baseline and counterfactual economies. Suppose worker x chooses occupation-hours pair $(j^B, \ell^B)_x$ in the baseline and $(j^{CF}, \ell^{CF})_x$ in the counterfactual. Realized utility in the baseline and counterfactual economies is then given by

$$U^{\omega} = (1 + \gamma_{ig}^{j^{\omega}})\nu_{g}^{j^{\omega}} c^{\omega} - \frac{(\ell^{\omega})^{\,1+\frac{1}{\xi}}}{1 + \frac{1}{\xi}}, \qquad \omega \in \{B, CF\},$$

where

$$c^{\omega} = \frac{1}{p^{\omega}}(1 - \tau_{g}^{j^{\omega}} - \kappa_{ig}^{j^{\omega},\omega})\, w_{ig}^{j^{\omega}} z_{ig}^{j^{\omega}} \varepsilon_{j^{\omega}}^{\mu} (1 + s^{\omega}) \ell^{\omega}, \qquad \omega \in \{B, CF\}.$$

For each individual x of type-subtype (i, g), welfare is given by the compensating variation ζ_x,

Table 9: Welfare effects of counterfactuals across groups (%)

Group	Undocumented → Documented	Non-fluent → Fluent	Recent (0–4 yrs) → Intermediate (5–10 yrs)	Combined legal, language, and tenure
Aggregate	0.38	0.19	0.01	0.90
Natives	-0.11	-1.05	-0.28	-0.93
Immigrants	2.43	5.49	1.20	8.72
Male	2.90	6.24	1.40	10.35
Female	1.99	4.79	1.02	7.19
Young	2.11	3.84	2.50	7.30
Prime	2.59	5.84	0.76	9.26
Old	2.57	6.76	0.41	9.54
No high school	4.42	7.02	0.34	12.44
High school	3.30	6.29	0.68	10.37
Some college	2.41	3.71	0.75	6.29
College and above	0.17	4.74	2.56	5.96
Documented	-0.13	4.40	0.89	5.25
Undocumented	7.16	7.49	1.76	15.11
Fluent	1.07	-1.17	1.18	1.05
Non-fluent	3.81	12.46	1.23	16.62
Recent	2.10	11.41	12.09	17.42
Intermediate	2.30	8.24	-0.29	9.87
Established	2.52	3.83	-0.29	6.94

Notes: This table reports average values, in percent, of the individual welfare measure $\zeta \times 100$ (as discussed in Section 5.3) by group under various counterfactuals. For each individual, ζ is the compensating variation that makes the individual indifferent between the baseline allocation and the counterfactual allocation. Specifically, it is the proportional change in baseline consumption needed to match utility under the counterfactual, holding fixed the individual's vector of idiosyncratic productivity draws $\{\epsilon_j\}_{j=0}^{j=J}$. The first row reports the aggregate average, followed by averages for natives, for immigrants, and for immigrant subgroups by sex, age, education, legal status, English ability, and time since arrival.

defined as the proportional change in baseline consumption that makes the individual indifferent between the baseline and counterfactual allocations. Let $U_x^B(\zeta_x)$ denote baseline utility when consumption is scaled by $1+\zeta_x$, and let U_x^{CF} denote counterfactual utility. Then ζ_x solves

$$U_x^B(\zeta_x) = (1+\gamma_{ig}^{j^B})\nu_g^{j^B}(1+\zeta_x)c^B - \frac{(\ell^B)^{1+\frac{1}{\xi}}}{1+\frac{1}{\xi}} = (1+\gamma_{ig}^{j^{CF}})\nu_g^{j^{CF}}c^{CF} - \frac{(\ell^{CF})^{1+\frac{1}{\xi}}}{1+\frac{1}{\xi}} = U_x^{CF}. \quad (8)$$

We then measure aggregate welfare as the economy-wide average of ζ_x. Group-specific welfare is computed analogously by averaging ζ_x within the relevant subgroup. Positive (negative) values of ζ_x indicate that the counterfactual raises (reduces) welfare relative to the baseline.[19]

Welfare results. Table 9 presents the welfare gains and losses of each counterfactual in the aggregate, for natives, and for immigrant groups. Aggregate welfare effects are positive but modest relative to the subgroup effects, with the joint counterfactual yielding the largest gain.

[19] An alternative metric is to compute a single ζ that would make a worker behind the veil of ignorance indifferent between the baseline and counterfactual economies. Results do not differ much across the two approaches.

Natives experience some welfare losses under all four counterfactuals, whereas immigrants experience larger positive average welfare gains in every case. This is consistent with the previous quantitative results, which show that employment, hours, and earnings responses are concentrated among immigrants, while native effects are negligible in magnitude.

We now turn to subgroups. The largest welfare gains from the legal-status counterfactual accrue overwhelmingly to undocumented immigrants, while documented immigrants are little affected. This aligns with the documented reallocation patterns, which showed that much of the adjustment under legalization is concentrated among certain immigrant groups. Similarly, the language counterfactual generates especially large welfare gains for non-fluent immigrants. This is consistent with the idea that language barriers are a key source of exclusion for non-fluent workers and that reducing them mainly benefits the subgroup most directly affected. Finally, the tenure counterfactual produces the largest gains for recent immigrants. This aligns with the previous results suggesting that easing labor-market integration for recent arrivals primarily benefits newer cohorts. Unlike the previous counterfactuals, in which welfare gains accrue disproportionately to less-skilled groups such as the less educated, the largest gains from this policy accrue to college-educated workers, with a much more even distribution across other dimensions such as legal status and fluency. Overall, the targeted counterfactuals differ in the breadth of welfare gains accruing to immigrants: legalization and language policies primarily benefit less-skilled groups, whereas tenure-related policies tend to benefit more-skilled groups.

Importantly, the joint counterfactual delivers the largest welfare gains overall for immigrants and for most immigrant subgroups, while native losses remain modest. This also reinforces earlier results where barriers along legal status, language, and tenure interact with one another, so addressing them jointly is much more effective than relaxing only one margin in isolation.

6 Discussion of Results

Finally, we explore the implications of alternative model specifications. For brevity, we focus on the main counterfactual in Section 5, which simultaneously reduces three dimensions of immigrant compensation wedges. We report the results in Table 10.

Modeling heterogeneity across occupations and workers. We examine the role of accounting for heterogeneity in occupations and worker types. To do so, we first estimate the model by classifying market occupations into four broad (task-based) occupations instead of 33. Output gains from reducing immigrant wedges are higher under this case. The rise in measured gains arise because hours now increase while it declines in the baseline. In the baseline model, reducing wedges reallocates workers across narrowly defined occupations, where increases in employment are driven by reallocation of less-productive workers who typically work fewer hours. When occupations are broadly classified, these reallocations are instead absorbed within the broad occupation groups. As a result, the adverse composition effect on hours is muted, making

the real GDP gains appear larger.

Next, we reduce the number of worker groups by distinguishing immigrants *only* by their time since arrival, and only consider subtypes of natives and immigrants by education. Thus, we are left with just 16 worker groups. Table 10 shows that gains from reducing wedges in this case are significantly reduced. This result is intuitive given that there is much less scope for misallocation due to wedges when the model does not sufficiently differentiate worker types.

Finally, we examine the role of distinguishing immigrants by legal status. By removing legal status as an immigrant characteristic, we are left with only 168 worker groups instead of 312 groups. Table 10 shows that, while real GDP gains are similar to those in the baseline, excluding legal status alters the sources of output gains from reducing immigrant barriers. In the baseline, gains are largely driven by increased employment, which is partly offset by a decline in hours while the change in TFP is negligible. In the absence of legal-status distinctions, changes in employment and hours are lower in magnitude, while TFP experiences modest gains.

In sum, accounting for rich heterogeneity across occupations and worker groups is important for the aggregate gains from reducing immigrant wedges and for their sources.

An alternative mapping of hourly wages in the model. In Section 3.3, we map hourly wages in the data to hourly wages in the model by measuring them as $(1-\tau_g^j-\kappa_{ig}^j)w_{ig}^j$. This assumption implies that annual hours worked in the model is $\ell z_{ig}^j \varepsilon_j$ and that $z_{ig}^j \varepsilon_j$ captures the ability to supply more hours or the ability to supply the same amount of hours at a lower disutility. To understand the role of this assumption, we consider an alternative mapping of hourly wages in the model such that hourly wages are given by $w_{ig}^j z_{ig}^j \varepsilon_j$. Table 10 shows that while TFP declines but hours increases in this setting relative to the baseline assumption. This is because, when we measure hourly wages as $w_{ig}^j z_{ig}^j \varepsilon_j$, reducing wedges does not directly affect wages, leading to smaller changes in employment and thus in output.[20]

Modeling endogenous labor supply. Next, we examine the role of elastic labor supply by considering a version of the model that abstracts from this channel. This exercise is also useful as it eliminates the need to differentiate between hourly wages and annual earnings in the model and tests how our assumption in Section 3.3 that the hourly wages in the model are $(1-\tau_g^j-\kappa_{ig}^j)w_{ig}^j$ affects our main results.[21] Table 10 shows that, in a model with inelastic labor supply (i.e., ℓ is exogenous), real GDP gains from reducing immigrant wedges drop to 0.28 percent because the decline in hours becomes larger. The larger decline in hours is driven by the absence of the endogenous increase in labor supply ℓ in the baseline model when wedges are reduced.

[20]Notice that in the alternative mapping, hours increase, which is the opposite of the baseline model's decomposition. This change is mechanical because the alternative mapping measures hours as labor supply ℓ so that changes in productivity composition, i.e., $z_{ig}^j \varepsilon_j$, now show up in TFP.

[21]In this case, however, we can only estimate productivities independent of occupations, i.e., z_{ig}.

Table 10: Gains from reducing immigrant wedges under alternative specifications

	Percent change				Change in
	Real GDP	TFP	Employment	Hours	immigrant share (pp)
Baseline	0.83	-0.00	1.31	-0.46	1.10
Fewer occupations	1.16	-0.15	1.08	0.23	0.91
Fewer worker groups	0.15	0.05	0.20	-0.10	0.18
No legal status	0.89	0.18	1.06	-0.35	0.90
Alternative mapping of hourly wages	0.27	-0.93	0.59	0.61	0.52
Inelastic labor supply	0.28	0.09	1.32	-1.11	1.13
Higher productivity draws for immigrants	0.83	-0.09	0.99	-0.06	0.84
Alternative legal-status classification	0.75	0.06	1.22	-0.53	1.04

Notes: This table presents the effects of the combined legal-status, English-proficiency, and tenure counterfactual under alternative model specifications. Real GDP, TFP, employment, and hours are reported as percent changes; the immigrant share is reported as a percentage-point change. Baseline refers to our baseline model; fewer occupations refers to an exercise where market occupations are grouped into four broad occupation categories; fewer worker groups refers to an exercise where we distinguish immigrants only by time since arrival, and only consider subtypes of natives and immigrants by education; no legal status refers to an exercise that shuts down legal-status distinctions; alternative mapping of hourly wages refers to a case where we measure hourly wages in the model as $w_{ig}^j z_{ig}^j \varepsilon_j$ instead of $(1-\tau_g^j-\kappa_{ig}^j)w_{ig}^j$; inelastic labor supply refers to when we shut down endogenous labor supply; higher productivity draws for immigrants refers to a model where the shape parameter of the Frechet distribution is different for immigrants such that the mean of productivity draws is 10 percent higher for immigrants than for natives; and alternative legal-status classification refers to a less restrictive rule for assigning documented status.

Heterogeneous productivity distributions between natives and immigrants. We assumed that idiosyncratic productivities of natives and immigrants across occupations are drawn from a common Frechet distribution, motivated by findings in Martellini et al. (2023). We now provide our results when we instead assume that productivities are drawn from different distributions. Specifically, we assume that the shape parameter of the Frechet distribution is different for immigrants such that the mean of productivity draws is 10 percent higher for immigrants. The reason why we focus on this case is that the estimation strategy of human capital gaps between natives and immigrants in Martellini et al. (2023) does not account for immigrant wedges when comparing wages between natives and immigrants. Absent wedges, their findings suggest a specification where productivities of natives and immigrants across occupations are drawn from a common distribution. Thus, accounting for wedges motivates this alternative specification.

Although the output gain is similar to the baseline, the sources of the gain differ in magnitude. When immigrants are assumed to draw productivities from a more favorable distribution, the decline in hours in the baseline model—which is driven by entry of low-productivity workers into employment who typically work fewer hours—becomes muted. As a result, the decline in hours in this case is much smaller. However, the gains in employment in this case are also smaller in magnitude. The lower increase in employment in this case is driven by a larger dispersion in idiosyncratic productivities of immigrants that leads to a lower mass of marginal workers.

Alternative legal-status classification. We also consider a less restrictive rule for identifying documented immigrants. As discussed in Section 3.1, this expands the documented group to include immigrants who receive any welfare benefits or are current graduate students. These criteria are motivated by the fact that undocumented immigrants are generally ineligible for

welfare and rarely pursue graduate education, making both characteristics informative of legal status. Table 10 shows that the gains from reducing immigrant wedges remain close to the baseline. Unlike the case where legal status is removed from the analysis altogether, the sources of the gains continue to resemble those in the baseline in magnitude. Thus, while the alternative classification lowers the estimated share of undocumented immigrants (and also slightly lowers the output gains), it does not largely alter the quantitative implications of reducing wedges.

Alternative parameter values. Finally, we examine our main findings under (*i*) alternative production technologies that differ in how labor bundles are aggregated (e.g., different nesting or different elasticities), and (*ii*) alternative values for other predetermined parameters. Table A5 in Appendix E summarizes our results. Overall, our main results are similar to our baseline with three intuitive exceptions: lowering substitutability between natives and immigrants, across different immigrant types, or in manual occupations leads to larger gains from reducing wedges.

7 Conclusion

We quantify the labor-market barriers faced by immigrants in the U.S. and assess their aggregate implications. We find that immigrant barriers are sizable and heterogeneous across occupations and immigrant groups. Using an occupational choice model disciplined by microdata, we show that these barriers distort the allocation of immigrant talent by preventing immigrants from working in the occupations in which they would be most productive. We estimate large immigrant-specific wedges, especially for recent, non-fluent, and undocumented immigrants, and conduct a series of validation exercises that support their interpretation as labor-market barriers.

We then study the aggregate and welfare effects of reducing these barriers. Removing immigrant barriers raises real GDP by about 1.5 percent, representing an upper-bound benchmark. Reducing wedges with more targeted policies aimed at legal status, language proficiency, and the integration of recent arrivals generate more modest gains when implemented individually. Nonetheless, a joint reduction in barriers along these margins yields output gains exceeding half of those under the upper-bound benchmark, suggesting that immigrant barriers reinforce one another. Overall, our results imply that policies toward immigrants should focus not only on entry flows, but also on how effectively immigrants are integrated into labor markets after arrival.

Our static framework abstracts from productivity dynamics and human-capital accumulation, which may interact with immigrant barriers. It also omits migration responses to these barriers, which may affect who migrates and thereby alter immigrant composition, the gains from reducing barriers, and the effects of immigration policies. We leave these dynamic and migration channels for future research.

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Online Appendix

A Model

In this section, we provide a formal definition of the equilibrium of the model.

Let each individual's idiosyncratic productivity vector be denoted by α, and let $\varphi(\alpha)$ denote the probability density function of individuals with vector α. Let the occupational choice of a type i, subtype g, and idiosyncratic productivity vector α be denoted by $\mathcal{O}_{ig}(\alpha) \in \{0, ..., J\}$.

A *competitive equilibrium* consists of prices $\left(p, \{p_j\}_{j=0}^J, \{w_{ig}^j\}_{i,g,j>0}, \{w_k^j\}_{k\in\{\text{nat,imm}\},j>0}, w^0\right)$ and allocations $\left(y, \{y_j\}_{j=0}^J, \{n_{ig}^j\}_{i,g,j>0}, \{n_k^j\}_{k\in\{\text{nat,imm}\},j>0}, n^0, \{\mathcal{O}_{ig}(\alpha), \ell_{ig}(\alpha)\}_{i,g}\right)$ such that:

1. Given price p and wages $\{w_{ig}^j\}_{j=1}^J$ and w^0, $\mathcal{O}_{ig}(\alpha)$ and $\ell_{ig}(\alpha)$ solve the problem of each individual of type i, subtype g, and productivity vector α.

2. Given price p_j and wages $\{w_k^j\}_k$, y_j and $\{n_k^j\}_k$ solve the problem of the representative firm in the outer nest of each market occupation $j = 1, ..., J$.

3. For each group $k \in \{\text{nat,imm}\}$, given wages w_k^j and $\{w_{ig}^j\}_{i\in\mathcal{I}_k,g}$, n_k^j and $\{n_{ig}^j\}_{i\in\mathcal{I}_k,g}$ solve the problem of the representative firm in the inner nest of each market occupation $j = 1, ..., J$.

4. Given price p_0 and wage w^0, y_0 and n^0 solve the problem of the representative firm in the non-market occupation.

5. Given prices p and $\{p_j\}_{j=0}^J$, y and $\{y_j\}_{j=0}^J$ solve the problem of the final good producer.

6. Aggregate revenue collected through compensation wedges is equal to aggregate reimbursements distributed to individuals:

$$\sum_{i=1}^{I}\sum_{g=1}^{G} N_{ig} \sum_{j=0}^{J} \int_\alpha (\tau_g^j + \kappa_{ig}^j) w_{ig}^j z_{ig}^j \varepsilon_j(\alpha) \ell_{ig}(\alpha) \mathbf{I}_{\{j=\mathcal{O}_{ig}(\alpha)\}} \varphi(\alpha) d\alpha$$
$$= \sum_{i=1}^{I}\sum_{g=1}^{G} N_{ig} \sum_{j=0}^{J} \int_\alpha s(1 - \tau_g^j - \kappa_{ig}^j) w_{ig}^j z_{ig}^j \varepsilon_j(\alpha) \ell_{ig}(\alpha) \mathbf{I}_{\{j=\mathcal{O}_{ig}(\alpha)\}} \varphi(\alpha) d\alpha.$$

7. Labor market clearing for individuals (i, g) in market occupation $j = 1, ..., J$ is:

$$n_{ig}^j = N_{ig} \times \int_\alpha z_{ig}^j \varepsilon_j(\alpha) \ell_{ig}(\alpha) \mathbf{I}_{\{j=\mathcal{O}_{ig}(\alpha)\}} \varphi(\alpha) d\alpha.$$

8. Labor market clearing in the non-market occupation is:

$$n^0 = \sum_{i=1}^{I}\sum_{g=1}^{G} \left(N_{ig} \times \int_\alpha z_{ig}^j \varepsilon_0(\alpha) \ell_{ig}(\alpha) \mathbf{I}_{\{0=\mathcal{O}_{ig}(\alpha)\}} \varphi(\alpha) d\alpha \right).$$

9. Market clearing of the final good is: $\sum_{i=1}^{I}\sum_{g=1}^{G}\int_{\alpha} c_{ig}(\alpha)\varphi(\alpha)d\alpha = y$.

For expositional simplicity, we do not use different notation to denote the demand and supply of occupation-specific goods. Thus, we abstract from the market clearing conditions for such goods, assuming that the same values that solve the problem of occupational goods producers also solve the problem of the final good producer.

B Data

This section provides details on the ACS, which underlies the empirical findings in Section 3.1. It also provides additional statistics on immigrants by demographics.

B.1 ACS

We use ACS 2010-2019 data to estimate the model for the U.S. In this section, we provide more details about the data, construction of variables, and measurement. In this data, we focus on a sample of non-business owners between the ages of 25 and 54 who are not in military.

For computing hourly wages, we require information on weeks worked during the year. We use weeks worked to convert annual earnings into hourly wages by dividing earnings by total annual hours, defined as weeks worked times usual hours worked per week. The ACS does not report the exact number of weeks worked for 2010–2018; for these years, only a binned version is available. We therefore impute weeks worked (`wkswork1`) using its binned counterpart, `wkswork2`. Between 2010–2018, for each bin, we assign the median number of weeks worked observed in 2019, the only year in which both the exact and binned measures are available.

The ACS provides information on individuals' citizenship and country of birth. The citizenship variable allows us to identify people who are not U.S. citizens or naturalized citizens, while the country of birth variable allows us to identify people born outside of the U.S. Using these variables, we define immigrants as foreign-born individuals who are either naturalized citizens or not citizens. This implies that natives' foreign-born children are classified as natives.

In our analysis, we consider an economy where immigrants are divided along various dimensions such as time since immigration, English fluency, and legal status. First, the ACS asks the year a foreign-born individual immigrated to the U.S. We use this information to classify immigrants into three groups by years since arrival: recent, with 0–4 years since arrival; intermediate, with 5–10 years since arrival; and established, with more than 10 years since arrival. Second, respondents also provide information on their English proficiency. We group immigrants into two groups: fluent immigrants, who speak only English or speak very well, non-fluent immigrants, who do not speak English very well. Finally, we classify immigrants by legal status as either documented or undocumented. We identify undocumented immigrants following Borjas (2017); details are provided in Section 3.1. In addition to these dimensions of heterogeneity, we also group immigrants and natives into subtypes based on their education, age, and gender.

Table A1: List of occupations

Non-routine cognitive	***Non-routine manual***
Management (10–430)	Healthcare support (3600–3650)
Business specialists (500–730)	Protective services (3700–3955)
Finance specialists (800–950)	Food services (4000–4150)
Computer: high (1005, 1010, 1020, 1030, 1106, 1200)	Cleaning and maintenance (4200–4250)
Computer: low (1000–1240, excl. high)	Personal care (4300–4650)
Architect and engineer (1300–1540)	***Routine manual***
Technicians (1550–1560)	Agriculture and forestry (6005–6130)
Science: low (1830–1980)	Construction (6200–6765)
Science: high (1600–1820)	Extraction (6800–6940)
Social services (2000–2060)	Installation: high (7000, 7030, 7140, 7200, 7220, 7240, 7330)
Legal: high (2100)	Installation: low (7000–7630, excl. high)
Legal: low (2105–2160)	Production (7700–8965)
Teachers (2200–2340)	Transportation: high (9000, 9030, 9040, 9110, 9200, 9300, 9310)
Teaching support (2400–2550)	Transportation: low (9000–9750, excl. high)
Arts and entertainment (2600–2920)	
Healthcare practitioners (3000–3250)	
Nurses and technicians (3255–3540)	
Routine cognitive	
Sales: high (4700, 4710, 4930)	
Sales: low (4700–4965, excl. high)	
Administrative support (5000–5940)	

Notes: This table presents the list of 33 market occupations included in our analysis. CPS occupation codes (`occ2010`) are shown in parentheses. For expositional purposes, some results in the paper are presented by grouping these occupations into four broad task-based categories: non-routine cognitive, routine cognitive, non-routine manual, and routine manual. The table lists the occupations included in each category.

We group occupations into 34 categories (33 market occupations and one non-market occupation). Our grouping of market occupations closely follows the two-digit 2010 SOC system, in which occupations are classified into 23 major groups. Starting from this classification, we further refine occupations in the following way. First, we do not include military-specific occupations, which are one of the occupation categories in the SOC system. Second, we separate business and financial operations occupations in the SOC system into two occupation categories (business operations and finance). Third, we separate technicians from architecture and engineering occupations. Fourth, we separate construction and extraction occupations into two occupation categories (construction and extraction). Finally, we split some broad occupation families into finer groups when there is likely meaningful heterogeneity in tasks, skill requirements, managerial and supervisory requirements, or institutional barriers within the underlying SOC category. In particular, we define "high" and "low" subgroups when deemed necessary. In the end, we arrive at 33 market occupations, as listed in Table A1.

While our estimation and results are based on these 33 market occupations, for expositional purposes, we often present results where we group these 33 market occupations into four task-based occupation categories. Following Autor and Dorn (2013), we group occupations along two dimensions of the characteristics of tasks required for the job: routine vs. non-routine and cognitive vs. manual. We then assign 33 market occupations into one of the four task-based occupation groups as in Cortes et al. (2020). Table A1 presents a list of 33 market occupations, their SOC codes, and their classification into four task-based occupation groups.

Given that we pool observations between 2010 and 2019 in our data, it is not common to

Table A2: Empirical moments for immigrants by demographics

	Distribution								
Occupation type	$I_{25–34}$	$I_{35–44}$	$I_{45–54}$	$I_{\text{No HS}}$	I_{HS}	$I_{\text{Some Col}}$	I_{Col}	I_m	I_f
Non-routine cognitive	0.23	0.25	0.23	0.02	0.06	0.19	0.57	0.26	0.22
Non-routine manual	0.15	0.16	0.17	0.22	0.23	0.18	0.05	0.15	0.17
Routine cognitive	0.12	0.11	0.12	0.05	0.13	0.19	0.11	0.10	0.13
Routine manual	0.19	0.21	0.22	0.36	0.29	0.17	0.05	0.35	0.08
Non-market	0.30	0.27	0.27	0.35	0.29	0.27	0.22	0.15	0.40
	Annual earnings								
Occupation type	$I_{25–34}$	$I_{35–44}$	$I_{45–54}$	$I_{\text{No HS}}$	I_{HS}	$I_{\text{Some Col}}$	I_{Col}	I_m	I_f
Non-routine cognitive	1.53	2.23	2.36	1.09	1.19	1.34	2.29	2.42	1.64
Non-routine manual	0.55	0.63	0.66	0.53	0.61	0.70	0.76	0.73	0.52
Routine cognitive	0.75	1.00	1.04	0.66	0.77	0.88	1.23	1.13	0.79
Routine manual	0.72	0.85	0.92	0.72	0.86	1.00	1.08	0.89	0.60
	Hourly wages								
Occupation type	$I_{25–34}$	$I_{35–44}$	$I_{45–54}$	$I_{\text{No HS}}$	I_{HS}	$I_{\text{Some Col}}$	I_{Col}	I_m	I_f
Non-routine cognitive	1.72	2.32	2.42	1.18	1.28	1.49	2.40	2.44	1.86
Non-routine manual	0.73	0.78	0.81	0.67	0.76	0.89	0.96	0.84	0.73
Routine cognitive	0.94	1.15	1.18	0.79	0.91	1.05	1.41	1.22	1.00
Routine manual	0.83	0.95	1.02	0.83	0.95	1.10	1.21	0.98	0.75

Notes: This table presents the distribution of immigrants across market and non-market occupations and their associated annual earnings and hourly wages using data from the 2010–2019 ACS. We first calculate the outcomes for each individual (type, subtype) pair in each occupation. For expositional purposes, we report the average moments for immigrants by age, education, and gender across four broad occupation categories, where we assign 33 market occupations into categories based on their skill and task-intensity: non-routine cognitive, non-routine manual, routine cognitive, and routine manual. The distribution of individuals across occupations is conditional on each worker group. Annual earnings and hourly wages are expressed relative to the respective values for native males ages 25 to 34 without a high school degree and employed in management occupations. $I_{25–34}$, $I_{35–44}$, and $I_{45–54}$ denote immigrants ages 25–34, 35–44, and 45–54, respectively. $I_{\text{No HS}}$, I_{HS}, $I_{\text{Some Col}}$, and I_{Col} denote immigrants with no high school degree, a high school degree, some college, and a college degree or more, respectively. I_m and I_f denote male and female immigrants, respectively.

have small cell sizes. However, when a cell has no observations, we assign an infinitesimally small mass of individuals, as well as infinitesimally small values for hourly wages and annual earnings, to that cell. Whenever a cell has at least one observation, we use information for these cells. Since what matters is the population share of these cells, which is small given the large number of cells, these cases do not significantly affect our results. We have experimented with different cutoff values higher than one observation and found similar results.

B.2 Immigrant demographics

In Section 3.1, Table 2 presents the distribution of individuals across market and non-market occupations and their associated annual earnings and hourly wages using data from the 2010–2019 ACS. There, immigrants are grouped by time since arrival, English proficiency, and legal status. Table A2 presents the same empirical moments when immigrants are instead grouped by demographics, namely education, age, and sex.

We note several key patterns. Younger and female immigrants are more likely to be in the non-market occupation. More educated immigrants are disproportionately represented in non-

routine cognitive occupations, while the less-educated are concentrated in routine manual and non-market activities. Male and female immigrants display similar distributions in non-routine cognitive, non-routine manual, and routine cognitive occupations. The main gender difference is in the residual allocation: male immigrants are more concentrated in routine manual jobs, whereas female immigrants are more concentrated in non-market activities. Finally, earnings and hourly wages also tend to be higher for older, more educated, and male immigrants.

C Estimation

This section provides derivations of model equations used in Section 3 when estimating the model. We then provide additional results in relation to our discussions in Section 3.

C.1 Derivations

We first present the derivation of Equations (2)-(7) in the paper.

Preliminaries. Our derivation of these equations relies on a few auxiliary results that are used throughout. The derivation of these auxiliary results is standard—for further details on some of these, see the appendix of Hsieh, Hurst, Jones, and Klenow (2019).

First, we have that the probability that workers (type, subtype) (i, g) choose occupation $j = 0, ..., J$ is given by:

$$\text{Fraction of employed}_{ig}^{j} = \frac{\left[(1+\gamma_{ig}^{j})\nu_{g}^{j}\left(1-\tau_{g}^{j}-\kappa_{ig}^{j}\right)w_{ig}^{j}z_{ig}^{j}\right]^{\eta}}{\sum_{q=0}^{J}\left[(1+\gamma_{ig}^{q})\nu_{g}^{q}\left(1-\tau_{g}^{q}-\kappa_{ig}^{q}\right)w_{ig}^{q}z_{ig}^{q}\right]^{\eta}}.$$

Second, we have that the geometric average earnings of a worker (type, subtype) (i, g) in occupation j is given by:

$$\begin{aligned}\text{Earnings}_{ig}^{j} = &\left[(1/p)\left(1+\gamma_{ig}^{j}\right)\nu_{g}^{j}\right]^{\xi}\left[\left(1-\tau_{g}^{j}-\kappa_{ig}^{j}\right)w_{ig}^{j}z_{ig}^{j}(1+s)\right]^{1+\xi} \\ &\times\left(\frac{1}{\text{Fraction of employed}_{ig}^{j}}\right)^{\frac{1+\xi}{\eta}}\exp\left[\frac{(1+\xi)\gamma_{em}}{\eta}\right],\end{aligned}$$

where γ_{em} is the Euler-Mascheroni constant.

Third, we have that the optimal labor demand in the inner nest of outer nest v in occupation j under perfect substitution is given by:

$$\sum_{i\in\mathcal{I}_v}\sum_{g=1}^{G}n_{ig}^{j} = \left(\frac{w_{v}^{j}}{p_j}\right)^{-\sigma_j}A_{j}^{\sigma_j-1}y_j.$$

Fourth, we have that the demand for the goods produced in occupation j is:

$$y_j = \left(\frac{p_j}{p}\right)^{-\sigma}y.$$

Finally, we have that the labor market clearing condition for workers (type, subtype) (i, g) in market occupation $j = 1, ..., J$ can be expressed as:

$$n_{ig}^{j} = N_{ig} z_{ig}^{j} \left[(1/p)(1 + \gamma_{ig}^{j} \nu_{g}^{j} \left(1 - \tau_{g}^{j} - \kappa_{ig}^{j}\right) w_{ig}^{j} z_{ig}^{j} (1+s) \right]^{\xi} \times \left(\text{Fraction of employed}_{ig}^{j}\right)^{\frac{\eta-(1+\xi)}{\eta}} \Gamma\left(1 - \frac{1+\xi}{\eta}\right).$$

Equation 1. Consider a pair (type, subtype) (i, g) and two alternative occupations j and k. The ratio of the geometric average earnings of these workers across the occupations is given by:

$$\frac{\text{Earnings}_{ig}^{j}}{\text{Earnings}_{ig}^{k}} = \frac{\left[\left(1 + \gamma_{ig}^{j}\right) \nu_{g}^{j}\right]^{\xi} \left[\left(1 - \tau_{g}^{j} - \kappa_{ig}^{j}\right) w_{ig}^{j} z_{ig}^{j}\right]^{1+\xi}}{\left[\left(1 + \gamma_{ig}^{k}\right) \nu_{g}^{k}\right]^{\xi} \left[\left(1 - \tau_{g}^{k} - \kappa_{ig}^{k}\right) w_{ig}^{k} z_{ig}^{k}\right]^{1+\xi}} \left(\frac{\text{Fraction of employed}_{ig}^{k}}{\text{Fraction of employed}_{ig}^{j}}\right)^{\frac{1+\xi}{\eta}}.$$

Plugging in the corresponding expressions for $\text{Fraction of employed}_{ig}^{j}$ and $\text{Fraction of employed}_{ig}^{k}$ and then simplifying, we obtain Equation (2):

$$\frac{\text{Earnings}_{ig}^{j}}{\text{Earnings}_{ig}^{k}} = \frac{(1 + \gamma_{ig}^{k})\nu_{g}^{k}}{(1 + \gamma_{ig}^{j})\nu_{g}^{j}}.$$

Equation 2. We derive Equation (4) by considering two worker (type, subtype) pairs (i, g) and (q, r) who choose a given occupation j:

$$\frac{\text{Earnings}_{ig}^{j}}{\text{Earnings}_{qr}^{j}} = \frac{\left[\left(1 + \gamma_{ig}^{j}\right) \nu_{g}^{j}\right]^{\xi} \left[\left(1 - \tau_{g}^{j} - \kappa_{ig}^{j}\right) w_{ig}^{j} z_{ig}^{j}\right]^{1+\xi}}{\left[\left(1 + \gamma_{qr}^{j}\right) \nu_{r}^{j}\right]^{\xi} \left[\left(1 - \tau_{r}^{j} - \kappa_{qr}^{j}\right) w_{qr}^{j} z_{qr}^{j}\right]^{1+\xi}} \left(\frac{p_{qr}^{j}}{\text{Fraction of employed}_{ig}^{j}}\right)^{\frac{1+\xi}{\eta}}.$$

We have that Equation (4) follows from setting $j = 0$ and (q, r) to base (type, subtype) pair (b, m) given that (i) wages are equated across worker types in the non-market occupation, (ii) immigrant compensation wedges and immigrant preference shifters are normalized to zero in occupation $j = 0$, (iii) common preferences are normalized to one and common compensation wedges are normalized to zero in occupation $j = 0$, (iv) productivity of base (type, subtype) in occupation $j = 0$ is normalized to one, and (v) earnings in the non-market occupation is set to a fraction λ of the weighted average of annual earnings across all market occupations for each (type, subtype).

Equation 3. We derive Equation (5) by considering a worker (type, subtype) (i, g) and two occupations j and k. We start with the relative allocations within workers across occupations:

$$\frac{\text{Fraction of employed}_{ig}^{j}}{\text{Fraction of employed}_{ig}^{k}} = \left[\frac{(1 + \gamma_{ig}^{j})\nu_{g}^{j}\text{Wages}_{ig}^{j} z_{ig}^{j}}{(1 + \gamma_{ig}^{k})\nu_{g}^{k}\text{Wages}_{ig}^{k} z_{ig}^{k}}\right]^{\eta}.$$

This expression follows from simplifying the ratio of probabilities $\text{Fraction of employed}_{ig}^{j}$ shown earlier in this section. Setting $k = 0$ and plugging Equation (2) we obtain the desired expression.

Equation 4. We derive Equation (6) by considering two outer nests v and q in a given occupation j. The derivation begins with obtaining an expression for relative wages w_{ig}^{j} as a function of observables and/or parameters that can be retrieved from observables up to this point. On

the one hand, we compute the relative demand for labor across workers within occupations:

$$\frac{\sum_{i\in\mathcal{I}_v}\sum_{g=1}^{G} n_{ig}^{j}}{\sum_{i\in\mathcal{I}_q}\sum_{g=1}^{G} n_{ig}^{j}} = \left(\frac{w_v^j}{w_q^j}\right)^{-\sigma_j}.$$

On the other hand, we use the market clearing conditions to compute the relative amount of labor across workers within occupations:

$$\frac{\sum_{i\in\mathcal{I}_v}\sum_{g=1}^{G} n_{ig}^{j}}{\sum_{i\in\mathcal{I}_q}\sum_{g=1}^{G} n_{ig}^{j}} = \left(\frac{w_v^j}{w_q^j}\right)^{\xi} \frac{\sum_{i\in\mathcal{I}_v}\sum_{g=1}^{G} N_{ig} z_{ig}^{j}\left[(1+\gamma_{ig}^{j})\nu_g^j\left(1-\tau_g^j-\kappa_{ig}^j\right)z_{ig}^j\right]^{\xi}\left(\text{Frac of emp}_{ig}^{j}\right)^{\frac{\eta-(1+\xi)}{\eta}}}{\sum_{i\in\mathcal{I}_q}\sum_{g=1}^{G} N_{ig} z_{ig}^{j}\left[(1+\gamma_{ig}^{j})\nu_g^j\left(1-\tau_g^j-\kappa_{ig}^j\right)z_{ig}^j\right]^{\xi}\left(\text{Frac of emp}_{ig}^{j}\right)^{\frac{\eta-(1+\xi)}{\eta}}}.$$

Equating the left-hand side of these expressions, we solve for relative wages:

$$\frac{w_v^j}{w_q^j} = \left\{\frac{\sum_{i\in\mathcal{I}_v}\sum_{g=1}^{G} N_{ig} z_{ig}^{j}\left[(1+\gamma_{ig}^{j})\nu_g^j \text{Wages}_{ig}^{j} z_{ig}^j\right]^{\xi}\left(\text{Fraction of employed}_{ig}^{j}\right)^{\frac{\eta-(1+\xi)}{\eta}}}{\sum_{i\in\mathcal{I}_q}\sum_{g=1}^{G} N_{ig} z_{ig}^{j}\left[(1+\gamma_{ig}^{j})\nu_g^j \text{Wages}_{ig}^{j} z_{ig}^j\right]^{\xi}\left(\text{Fraction of employed}_{ig}^{j}\right)^{\frac{\eta-(1+\xi)}{\eta}}}\right\}^{\frac{-1}{\sigma_j}}.$$

We plug this into the following expression that characterizes ratio of observed hourly wages:

$$\frac{\text{Wages}_{ig}^{j}}{\text{Wages}_{qr}^{j}} = \frac{\left(1-\tau_g^j-\kappa_{ig}^j\right)w_{ig}^j}{\left(1-\tau_r^j-\kappa_{qr}^j\right)w_{qr}^j}.$$

Equation (6) results from combining these expressions where we set $q = 1$ (i.e., natives) which implies setting $\kappa_{qr}^j = 0$ and dropping summations over $i \in \mathcal{I}_q$ as there is only one native type (all natives).

Equation 5. Consider now the relative demand for labor across market occupations j and k within outer nest v:

$$\frac{\sum_{i\in\mathcal{I}_v}\sum_{g=1}^{G} n_{ig}^{j}}{\sum_{i\in\mathcal{I}_v}\sum_{g=1}^{G} n_{ig}^{k}} = \frac{\left(\frac{w_v^j}{p_j}\right)^{-\sigma_j} A_j^{\sigma_j-1} y_j}{\left(\frac{w_v^k}{p_k}\right)^{-\sigma_k} A_k^{\sigma_k-1} y_k}.$$

Plugging in the solution to the final good producer's problem, we obtain:

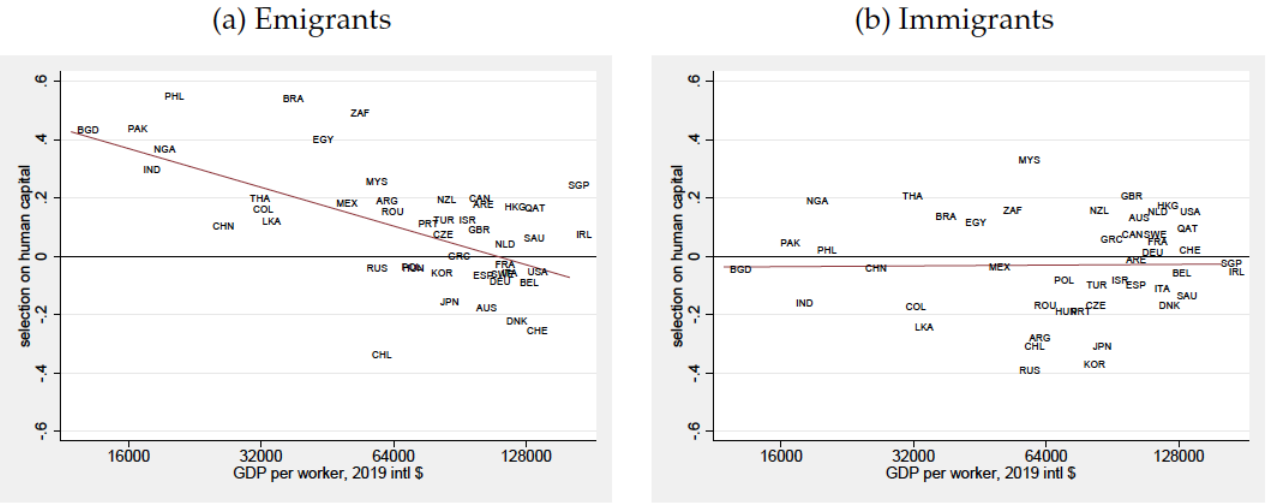
$$\frac{\sum_{i\in\mathcal{I}_v}\sum_{g=1}^{G} n_{ig}^{j}}{\sum_{i\in\mathcal{I}_v}\sum_{g=1}^{G} n_{ig}^{k}} = \frac{\left(\frac{w_v^j}{p_j}\right)^{-\sigma_j} A_j^{\sigma_j-1}}{\left(\frac{w_v^k}{p_k}\right)^{-\sigma_k} A_k^{\sigma_k-1}}\left(\frac{p_j}{p_k}\right)^{-\sigma}.$$

Let $\sigma_j = \sigma_k = \sigma$ for all j and k. Then, the expression can be simplified to obtain:

$$A_j = \left\{\left(\frac{w_v^j}{w_v^k}\right)^{\sigma} A_k^{\sigma-1}\frac{\sum_{i\in\mathcal{I}_v}\sum_{g=1}^{G} n_{ig}^{j}}{\sum_{i\in\mathcal{I}_v}\sum_{g=1}^{G} n_{ig}^{k}}\right\}^{\frac{1}{\sigma-1}}.$$

Then, we obtain Equation (7) by implementing the following steps: (i) plug in the respective labor market clearing conditions, (ii) substitute the wage ratio used in the last step of the

Figure A1: Average human capital of emigrants and immigrants relative to non-migrants



Source: Martellini, Schoellman, and Sockin (2023)

Notes: This figure is copied from Martellini, Schoellman, and Sockin (2023) to provide ease of reference. For each country c, Panel (a) shows the log difference in average human capital for emigrants from c as compared to non-migrants. Panel (b) shows the same for immigrants to c relative to non-migrants. Countries are ordered in PPP GDP per worker (in log scale) in horizontal axis of both figures. For more details, please refer to Martellini, Schoellman, and Sockin (2023).

derivation of Equation (6), (*iii*) set $k = 1$ with normalization $A_1 = 1$, and (*iv*) write the equation for base (type, subtype) (b, m) (and dropping summations over $i \in \mathcal{I}_v$ as the base type is natives and there is only one native type) and simplify.

C.2 Additional estimation results

Average human capital across immigrants, emigrants, and non-migrants. In Section 3.2, we assume that the shape parameter η_i of the Frechet distribution of idiosyncratic productivities is the same for natives and immigrants. This assumption implies that immigrants and natives draw η_i from the same distribution.

This assumption is motivated by the recent evidence from Martellini, Schoellman, and Sockin (2023) on the average human capital of emigrants, immigrants, and non-migrants across countries. Panel (a) of Figure 4 of their paper (reproduced in Figure A1 for ease of reference) shows the log difference in average human capital for emigrants from c as compared to non-migrants. Panel (b) shows the same for immigrants to c relative to non-migrants. Countries are ordered in PPP GDP per worker (in log scale) in horizontal axis of both figures.

Panel (a) shows that emigrants are more positively selected when migrating from less-developed economies. Thus, despite significant differences in the quality of education between rich and poor countries, the average human capital of immigrants in destination countries is close to that of natives (Panel (b)). Motivated by these findings, and given that we do not model migration decisions, we assume in our baseline calibration that the shape parameter η_i of the Frechet distribution of idiosyncratic productivities is the same for natives and immigrants.

Table A3: Estimation results for distribution, annual earnings, and hourly wages

	Distribution							
Occupation type	N	I_{0-4}	I_{5-10}	I_{10+}	$I_{\text{Low Eng}}$	$I_{\text{High Eng}}$	I_{Undoc}	I_{Doc}
Non-routine cognitive	0.32	0.28	0.25	0.25	0.11	0.40	0.17	0.30
Non-routine manual	0.11	0.16	0.21	0.18	0.23	0.13	0.21	0.16
Routine cognitive	0.19	0.11	0.12	0.14	0.10	0.16	0.10	0.15
Routine manual	0.16	0.17	0.22	0.24	0.31	0.14	0.29	0.19
Non-market	0.22	0.28	0.21	0.20	0.25	0.17	0.22	0.21
	Annual earnings							
Occupation type	N	I_{0-4}	I_{5-10}	I_{10+}	$I_{\text{Low Eng}}$	$I_{\text{High Eng}}$	I_{Undoc}	I_{Doc}
Non-routine cognitive	1.76	1.76	2.04	2.16	1.59	2.21	2.19	2.05
Non-routine manual	0.75	0.49	0.60	0.68	0.57	0.76	0.58	0.68
Routine cognitive	1.03	0.76	0.86	1.01	0.77	1.08	0.87	1.00
Routine manual	1.07	0.68	0.79	0.93	0.79	1.08	0.77	0.97
	Hourly wages							
Occupation type	N	I_{0-4}	I_{5-10}	I_{10+}	$I_{\text{Low Eng}}$	$I_{\text{High Eng}}$	I_{Undoc}	I_{Doc}
Non-routine cognitive	1.68	1.86	1.98	2.05	1.64	2.11	2.11	1.98
Non-routine manual	0.86	0.69	0.71	0.78	0.69	0.87	0.68	0.81
Routine cognitive	1.09	0.97	0.97	1.07	0.88	1.15	0.97	1.08
Routine manual	1.08	0.81	0.82	0.94	0.82	1.08	0.80	0.99

Notes: This table presents model-implied targeted moments for the allocation of individual types as well as their annual earnings and hourly wages across occupations. We first calculate the outcomes for each individual (type, subtype) pair in each occupation. For expositional purposes, we report the average moments for natives and immigrant types across four broad occupation categories, where we assign 33 market occupations into categories based on their skill and task-intensity: non-routine cognitive, non-routine manual, routine cognitive, and routine manual. The distribution of individuals across occupations is conditional on each worker type. Annual earnings and hourly wages are expressed relative to respective values for native males of ages 25 to 34 without high school degree and employed in management occupations. N denotes natives, I_{0-4} denotes recent immigrants ($\leq$ 4 years), I_{5-10} denotes intermediate immigrants (5–10 years), I_{10+} denotes established immigrants ($>$ 10 years), $I_{\text{Low Eng}}$ denotes low English proficiency immigrants, $I_{\text{High Eng}}$ denotes high English proficiency immigrants, I_{Undoc} denotes undocumented immigrants, and I_{Doc} denotes documented immigrants.

Estimation results. Table A3 shows the model counterparts of the empirical moments presented in Table 2. Overall, the model closely matches the empirical moments in Table 2.

D Additional Results on Counterfactuals

In this section, we provide additional results to complement our discussions in Section 5.

Role of within-market reallocation. Table 7 in Section 5.2 documents the aggregate gains from reducing immigrant wedges. These gains are obtained due to both flows of non-employed individuals from the non-market occupation to market occupations and flows of employed workers within market occupations. To quantify the role of flows of individuals between the non-market occupation and market occupations, Table A4 recomputes the effects of each counterfactual when we prevent individuals from moving in and out of the non-market occupation. A discussion of results from Table A4 are already provided in the main text.

Table A4: Occupation-level effects across counterfactuals: within-market reallocation only

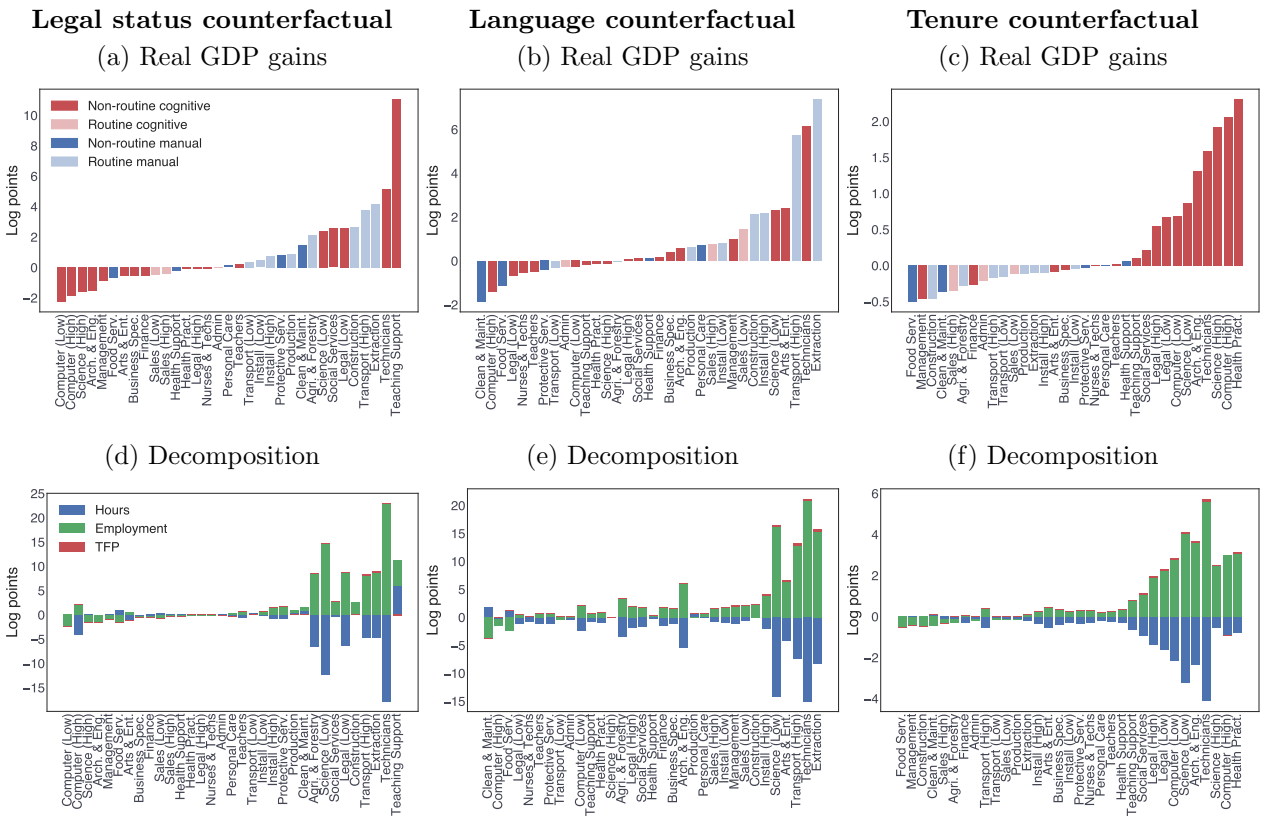
	Percent change				Change in
Occupation type	Real GDP	TFP	Employment	Hours	immigrant share (pp)
Undocumented → Documented					
Aggregate	-0.13	-0.16	0.00	0.04	0.00
Non-routine cognitive	-0.41	-0.22	0.23	-0.42	0.20
Non-routine manual	0.00	0.10	-0.74	0.65	-0.54
Routine cognitive	-0.33	-0.07	-0.44	0.18	-0.38
Routine manual	0.91	0.07	0.58	0.25	0.43
Non-market	-0.04	0.03	0.00	-0.06	0.00
Non-fluent English → Fluent English					
Aggregate	-0.05	0.19	0.00	-0.24	0.00
Non-routine cognitive	-0.03	0.11	1.04	-1.17	0.87
Non-routine manual	-1.19	0.15	-2.47	1.15	-1.83
Routine cognitive	0.07	0.19	0.17	-0.29	0.14
Routine manual	0.37	0.20	-0.25	0.42	-0.19
Non-market	-0.61	0.13	0.00	-0.74	0.00
Recent (0–4 years) → Intermediate (5–10 years)					
Aggregate	-0.09	0.06	0.00	-0.15	0.00
Non-routine cognitive	0.09	0.05	0.39	-0.35	0.32
Non-routine manual	-0.28	0.05	-0.35	0.02	-0.25
Routine cognitive	-0.27	-0.00	-0.18	-0.08	-0.16
Routine manual	-0.29	0.00	-0.26	-0.03	-0.19
Non-market	-0.16	0.04	0.00	-0.19	0.00
Combined legal, language, and tenure adjustment					
Aggregate	0.23	0.00	0.00	0.23	0.00
Non-routine cognitive	-0.18	-0.02	0.60	-0.75	0.50
Non-routine manual	-0.32	0.15	-2.34	1.92	-1.73
Routine cognitive	0.07	0.17	-0.15	0.06	-0.13
Routine manual	1.99	0.27	0.80	0.91	0.59
Non-market	-0.46	0.17	0.00	-0.63	0.00

Notes: This table presents the aggregate and occupation-level effects of each decomposition counterfactual on real GDP, TFP, employment, hours, and the immigrant share. The first four columns report percent changes relative to the baseline, while the final column reports percentage-point changes in the immigrant share. Aggregate real GDP is output produced in the market sector, total factor productivity (TFP) is real GDP per hour, employment is the mass of workers in market occupations (or each occupation), and hours is the average hours worked in market occupations (or each occupation). The change in the immigrant share denotes the percentage point (pp) change in the fraction of immigrants employed in market occupations or each occupation. Unlike the corresponding standard counterfactuals, these decomposition exercises prevent inflows to and outflows from the non-market occupation when immigrant wedges are removed, thereby isolating the effects of within-market reallocation. For expositional purposes, we group the 33 market occupations into four broad categories based on skill and task intensity: non-routine cognitive, non-routine manual, routine cognitive, and routine manual. Non-market refers to the home sector.

Effects of reducing immigrant wedges across all occupations. Table 7 in Section 5.2 analyzes the effects of reducing immigrant wedges across occupations, where we grouped occupations into four broad task-based occupation categories for expositional purposes. Figure A2 presents results across all 33 occupations in our analysis.

Panels (a) and (b) show that, under the legal-status and language counterfactuals, routine

Figure A2: Real GDP gains and their sources across market occupations

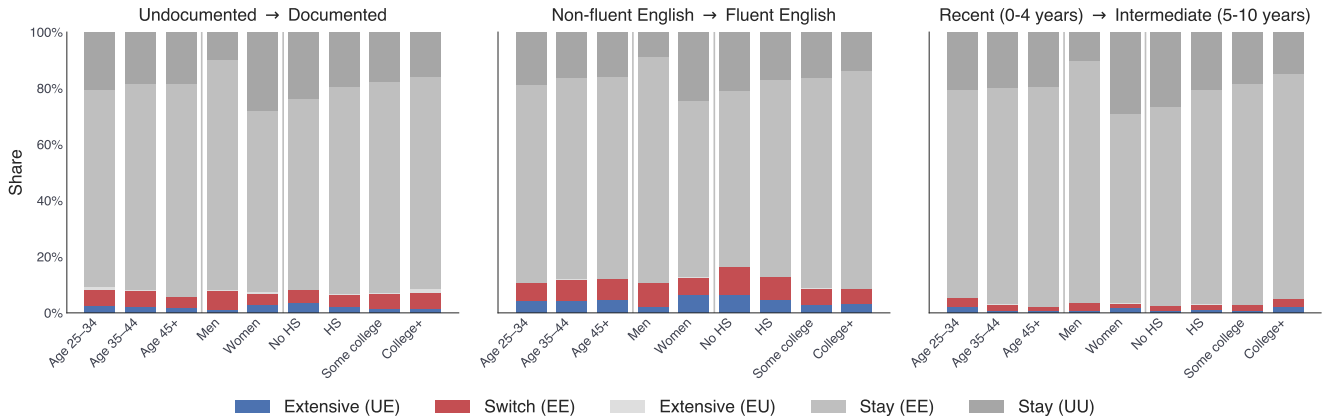


Notes: The columns correspond to three counterfactuals: undocumented → documented, non-fluent English → fluent English, and recent (0–4 years) → intermediate (5–10 years). The top row plots occupation-level real GDP gains across all market occupations. The bottom row decomposes occupation-level real GDP gains into the contributions of TFP, employment, and average hours. In each column, occupations are ordered from the smallest to the largest real GDP gain under the corresponding counterfactual. Real GDP gains and their components are reported in log points.

manual occupations such as extraction, transportation (high), and construction experience the largest output gains. While several non-routine cognitive occupations, such as technicians and lower-skill science-related jobs, also experience output gains, many others experience output losses, offsetting aggregate gains. In contrast, Panel (c) shows that the tenure counterfactual yields its largest gains in non-routine cognitive jobs. Across all counterfactuals, most output gains come from increases in employment, often partly offset by declines in hours while changes in TFP are smaller in magnitude (Panels (d) through (f)).

Reallocation patterns across immigrants by demographics. Figure 3 in Section 5.2 presents the distribution of reallocation types across immigrant subgroups defined by legal status, English fluency, and tenure under each counterfactual. For each subgroup, the figure reports the fraction of workers making each of five possible transitions: movements from the non-market occupation to market occupations (Extensive UE), switches between market occupations conditional on being employed in both the baseline and counterfactual economies (market-to-market switching), movements from market occupations to the non-market occupation (Extensive EU), remaining in the same market occupation (Stayer EE), and remaining in the same non-market

Figure A3: Decomposition of transition responses by immigrant demographics



Notes: This figure plots the decomposition of transition responses across immigrant demographics for three counterfactuals: undocumented → documented, non-fluent English → fluent English, and recent (0–4 years) → intermediate (5–10 years). Within each panel, bars report the share of the total transition response accounted for by five components: extensive margin from non-employment to employment (UE), market-to-market occupation switching (EE switch), extensive margin from employment to non-employment (EU), stayers in employment and the same occupation (EE stay), and stayers in non-employment (UU). Each bar sums to one. Subgroups are organized by age, sex, and educational attainment. The blue and red segments highlight the UE and the EE components, while the gray segments denote the remaining adjustment channels.

occupation (Stayer UU). Figure A3 presents the same decomposition, but groups immigrants instead by demographic characteristics such as age, gender, and educational attainment.

Overall, under the legal-status and language counterfactuals, most extensive-margin (non-market to market) and market-to-market switches occur among less-educated immigrants. By contrast, the educational patterns differ under the tenure counterfactual, where more reallocation occurs among educated immigrants, although small in magnitude.

E Results under Alternative Parametrizations

In this section, we present our main results on changes in aggregate real GDP, TFP, employment, and hours worked when we simultaneously equalize three dimensions of immigrant compensation wedges—legal status, English fluency, and tenure—under alternative parametrizations of our baseline model, as discussed in Section 6. We consider (i) alternative production technologies that differ in how labor bundles are aggregated across worker types and subtypes (e.g., different nesting, as well as different elasticities), and (ii) alternative values for other predetermined parameters. In each of these cases, we re-estimate the model's parameters and wedges and then compute changes in outcomes when immigrant wedges are reduced. These results are summarized in Table A5. Overall, our main results are similar to our baseline results with three intuitive exceptions: lowering substitutability between natives and immigrants, across different immigrant types, or in manual occupations leads to larger gains from reducing wedges.

First, in Section 3.2, following Ottaviano and Peri (2012), we set the elasticity of substitution between natives and immigrants in the outer nest to $\sigma_j = 20 \quad \forall j = 1, ..., J$. While this is their preferred estimate when the native-immigrant elasticity is restricted to be the same for all education groups as in our baseline estimation, we acknowledge that there are alternative

Table A5: Gains from reducing immigrant wedges under alternative parametrizations

	Percent change				Change in
	Real GDP	TFP	Employment	Hours	immigrant share (pp)
Baseline	0.83	-0.00	1.31	-0.46	1.10
Elasticity of substitution between natives and immigrants with $\sigma_j = 4.6$	1.15	0.23	1.15	-0.23	1.09
Imperfect substitution across natives and all immigrant types	1.49	0.36	1.55	-0.42	1.32
Perfect substitution across natives and all immigrant types	0.79	-0.04	1.33	-0.50	1.10
Imperfect substitution between education groups	0.85	-0.04	1.32	-0.43	1.11
Elasticity of substitution for manual occupations, $\sigma_j = 4.6$	1.10	0.05	1.54	-0.48	1.20
Elasticity of substitution for manual occupations, $\sigma_j = 12.5$	0.91	-0.00	1.40	-0.49	1.16
Higher value for elasticity of substitution across subtypes in inner nest	0.93	-0.04	1.51	-0.53	1.27
25 percent UI replacement rate, $\lambda = 0.25$	0.77	0.00	1.31	-0.53	1.10
75 percent UI replacement rate, $\lambda = 0.75$	0.89	-0.01	1.31	-0.40	1.10

Notes: This table presents the effects of the combined legal-status, English-proficiency, and tenure counterfactual under alternative parameter values. Real GDP, TFP, employment, and hours are reported as percent changes; the immigrant share is reported as a percentage-point change. Please refer to main text for a detailed discussion on these exercises.

values used across different studies. For this reason, we present our main results under $\sigma_j = 4.6 \ \ \forall j = 1, ..., J$ as in Burstein, Hanson, Tian, and Vogel (2020). Intuitively, when immigrant and native labor bundles are much less substitutable, real GDP gains from reducing immigrant wedges becomes much larger, increasing to 1.15 percent from its baseline value of 0.83 percent.

Second, in our model, we assume that the outer nest aggregates labor bundles of natives and all types of immigrants (without taking into account different immigrant types). Here, we make a change to the production technology so that the outer nest aggregates worker bundles of natives and all 12 types of immigrants (i.e., an aggregation across 13 worker bundles instead of 2 in the baseline specification). Recall that, in Section 3.3, we assume that labor bundles in the outer nest are imperfect substitutes, while labor bundles in the inner nest are perfect substitutes. Thus, the implication of this change in the production technology is that immigrants of different types now become imperfectly substitutable. This captures the possibility that immigrants with different characteristics based on time since arrival, fluency in English, and legal status may be imperfectly substitutable. As Table A5 shows, when these types of immigrants are imperfect substitutes, real GDP gains from reducing immigrant wedges are larger at 1.49 percent. This exercise shows that our baseline specification where all immigrant types are perfect substitutes sets a lower bar for gains from reducing wedges. When all types of immigrants are imperfect substitutes, our framework predicts larger gains from reducing wedges.

Third, we consider perfect substitution between labor bundles in the outer nest that aggregates labor bundles of natives and all types of immigrants. We approximate perfect substitution in the outer nest with $\sigma_j = 40 \ \ \forall j = 1, ..., J$, the same value we use to approximate perfect substitution in the inner nest with $\widetilde{\sigma}_j = 40 \ \forall j = 1, ..., J$. Since our baseline of $\sigma_j = 20 \ \ \forall j = 1, ..., J$ already assumes a large degree of substitutability across immigrant and native labor bundles in the outer nest, our results do not significantly change when we instead assume $\sigma_j = 40 \ \ \forall j = 1, ..., J$.

Fourth, we consider a production structure in which individuals with different education levels are imperfect substitutes. Specifically, the outer nest aggregates worker bundles between college

natives, non-college natives, college immigrants, and non-college immigrants. While we do not allow the elasticity parameter itself to vary by education or experience, as in Ottaviano and Peri (2012), this richer education-based nesting structure allows substitution patterns to differ across broad skill groups while maintaining a common elasticity parameter within the nest. We find that this change in the production technology does not materially alter our main results.

Fifth, we consider a case in which the elasticity of substitution between natives and immigrants varies across occupation groups. This is motivated by Ottaviano and Peri (2012), who find lower substitutability between natives and immigrants for less-educated workers, but no evidence of imperfect substitutability for college-educated workers. To capture this idea in a tractable way, we let σ_j differ between cognitive and manual occupations, where cognitive occupations capture jobs closely associated with college-educated labor, and manual occupations as those more closely associated with non-college labor. Accordingly, we keep $\sigma_j = 20$ for cognitive occupations, as in the baseline, and set $\sigma_j = 4.6$ for manual occupations. As Table A5 shows, this raises the gains from reducing immigrant wedges relative to the baseline. Intuitively, lower substitutability in manual occupations makes distortions in those occupations more costly, amplifying the aggregate effects of reducing immigrant wedges, especially considering that the largest wedges are concentrated among manual jobs as documented in Section 4.

Sixth, we consider the same exercise, except that we set the elasticity of substitution in manual occupations to $\sigma_j = 12.5$, motivated by Ottaviano and Peri (2012), who estimate an average elasticity for groups *up to* some college education. We find that the gains from reducing immigrant wedges remain larger than in the baseline, but smaller than in the case where manual occupations are even less substitutable, with $\sigma_j = 4.6$.

Seventh, in our estimation, we approximate the perfect substitution across labor bundles in the inner nest with $\widetilde{\sigma}_j = 40 \ \forall j = 1, ..., J$. Table A5 shows that our results remain similar under an even higher value of this elasticity (i.e., $\widetilde{\sigma}_j = 80 \ \forall j = 1, ..., J$).

Finally, in our baseline calibration, for each individual (type, subtype) pair, we set annual earnings in the non-market occupation to 50 percent of the weighted average of annual earnings across market occupations, i.e., $\lambda = 0.5$. Using alternative values, $\lambda = 0.25$ or $\lambda = 0.75$, does not materially alter our results.